

State-dependent Distribution Friction and the Transmission of Monetary Policy*

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Abstract

We examine the role of state-dependent distribution frictions in the transmission of monetary policy shocks in a dynamic multi-sector general equilibrium model. Our benchmark model predicts a counter-cyclical distribution margin, which is consistent with micro price evidence. We then estimate an augmented model with price and wage rigidities and attribute the lion's share of this countercyclicality to service-sector productivity shocks. By allowing for distribution frictions to depend on the state of the economy, which substantially alters monetary transmission, we find that state dependence weakens the real effects of monetary policy by dampening the consumption response to monetary shocks. In stark contrast, such a result cannot be replicated when distribution frictions are time-invariant. Our findings highlight a novel channel through which the interplay between sectoral productivity and distribution frictions jointly influences monetary policy transmission.

JEL Codes: E3, E5

Keywords: State-dependent Distribution Friction, Distribution Margin, Monetary Policy, Nominal Rigidities.

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1. Introduction

The shortage of truck drivers and severe port congestion in mid-2021 intensified supply chain disruptions in the United States, and the resulting economic strains caused by the COVID-19 pandemic. Rattled consumers faced empty store shelves and prolonged delivery delays, particularly for electronics and furniture, at a time when prices were rising rapidly. These pandemic-era disruptions in the distribution sector had sizable effects on goods prices and consumption, highlighting the macroeconomic importance of the distribution service.

The distribution sector indeed plays an important role in modern economies. Distribution costs account for over 40% of the retail price of goods in the United States ([Berger et al., 2012](#); [Crucini and Landry, 2019](#)) and around 60% in Argentina ([Burstein et al., 2003](#)). Even though distribution frictions are central to understanding price dynamics, such as exchange-rate pass-through to import prices ([Corsetti and Dedola, 2005](#); [Choudhri et al., 2005](#); [Campa and Goldberg, 2010](#)), the literature has largely remained silent on the cyclical behavior of this price wedge and its implications for monetary policy transmission. Addressing this gap is essential for understanding the real effects of monetary policy shocks, given the quantitative importance of distribution frictions for prices and, consequently, for consumption. In this paper, we examine the role of state-dependent distribution frictions (i.e., the physical quantity of aggregate services required to deliver one unit of manufactured goods) in the transmission of monetary policies.

To that end, we consider a dynamic multi-sector model that is disciplined by a separate distribution sector where distribution friction varies with the state of the economy. The benchmark model predicts a counter-cyclical distribution margin, a pattern of business-cycle co-movement that is consistent with micro-level price data from Brazil. More importantly, we find that state-dependent distribution friction dampens consumption responses to monetary shocks. A model with time-invariant distribution friction fails to replicate this result, underestimating the quantitative significance of the distribution friction.

Turning to more details, our analysis begins with a benchmark dynamic multi-sector model à la [Corsetti and Dedola \(2005\)](#) to explore the cyclical behavior of the distribution margin. The economic environment features two production sectors: manufacturing and services. Manu-

facturing relies on the distribution sector to deliver goods from producers to retailers, while the distribution sector, in turn, uses differentiated services as intermediate inputs. We model distribution friction as the aggregated service units needed to deliver a manufactured good,¹ introducing state-dependent friction, unlike existing studies that assume time-invariant friction. Motivated by recent research emphasizing the overlooked significance of internal trade barriers (Van Leemput, 2021), we construct a closed-economy variant of the model proposed by Corsetti and Dedola (2005) that centers on internal distribution frictions. In our benchmark model featuring flexible prices and wages, we analytically show that the distribution margin is counter-cyclical in equilibrium.

To improve clarity, it is helpful to distinguish several closely related concepts concerning distribution friction using the key notations employed in our model. Specifically, we follow Burstein et al. (2003) and define the distribution margin as the ratio of distribution cost to the retail price of manufactured goods: $\delta_t = (\eta_t P_{S,t})/p_t(m)$, where δ_t is the distribution margin, η_t is the distribution friction, and $P_{S,t}$ and $p_t(m)$ are the retail prices of services and manufactured goods, respectively. The multiplicative term $\eta_t P_{S,t}$ thus represents the distribution cost. In prior work, such as Corsetti and Dedola (2005), the distribution margin and distribution cost are treated as state-dependent through their relationship with retail prices, but the distribution friction η_t itself is assumed to be constant. In contrast, we introduce state-dependent distribution friction, modeled as $\eta_t \equiv \eta(\Omega_t)$, where Ω_t reflects the state of the economy. We show that such a generalization has important macroeconomic implications, in particular by revealing the counter-cyclical nature of the distribution margin and clarifying the transmission of monetary policy through the distribution-related channel.

We use novel microdata on Brazilian fuel prices to verify the countercyclicality of the distribution margin as predicted by the benchmark model. Our dataset includes various price margins, such as retail price and distribution cost for gas stations across Brazil. Notably, our data covers different fuel types at the goods level. Unlike Peter and Ruane (2022), who use a static model with Indian manufacturing data, our dynamic model captures the distributional

¹See, for example, Burstein et al. (2003), Corsetti and Dedola (2005), Bergin and Corsetti (2023), and Corsetti et al. (2023) for similar modeling strategies. See Bodenstein and Corsetti (2025) for a survey of international monetary policy transmission mechanisms involving distribution costs.

wedge's cyclicality and emphasizes the real effects of monetary policies. A potential concern is bias introduced by the use of fuel price data, since fuel is part of the distribution cost. However, distribution costs also include substantial rental and depreciation expenses for transport equipment, warehouse setup costs, and labor costs. The low correlation (0.06) between distribution costs and gasoline prices in our microdata further suggests that fuel is unlikely to be the dominant component.

Our sample includes average price margins for retail prices and distribution costs across gas stations in each Brazilian state, recorded weekly and merged with aggregate state- and national-level variables. We regress the distribution margin on various economic indicators. The results show a negative association between the distribution margin and output growth, consistent with our benchmark model's prediction. Although based on fuel-sector data, this finding has significant implications for the broader Brazilian economy, as the fuel sector accounts for approximately 11% of the economy and serves as a crucial intermediate input for other sectors. Additionally, we cross-validate this countercyclical relationship using historical data on U.S. farm food products and survey data on manufacturing firms in India.

Motivated by the observed countercyclical relationship, we further assess the quantitative importance of state-dependent distribution friction in an augmented version of the benchmark model with price and wage rigidities. This model involves retailing firms that produce and sell manufactured goods directly to consumers, internalizing production and delivery services. As in the benchmark, goods are distributed through a sector that relies on service-sector inputs, and distribution friction varies with the state of the economy. Retailers' pricing behavior follows [Calvo \(1983\)](#), where prices change each period with a predetermined probability.

Using Brazilian aggregate data, we estimate an augmented model to quantify the impact of state-dependent distribution friction on the transmission of monetary policy shocks. Following the extensive literature on the real effects of money,² we introduce monetary policy shocks as innovations to the money supply in the baseline augmented model and later as exogenous innovations of a policy rule akin to [Taylor \(1993\)](#) in a sensitivity analysis. Before adopting the augmented model as a workhorse to explore the quantitative importance of state-dependent

²See, for example, [Christiano et al. \(1999\)](#), [Golosov and Lucas Jr. \(2007\)](#), and [Nakamura and Steinsson \(2010\)](#).

distribution friction for the transmission of monetary policies, we show in a model fit check that our preferred model, which estimates the size parameter governing the magnitude of the distribution friction and calibrates the curvature and state-dependence parameters in the distribution friction function, matches Brazilian data significantly better than alternative models.

A key finding from our preferred quantitative model is that state-dependent distribution friction significantly dampens the real effects of expansionary monetary policy shocks by suppressing consumption responses. The augmented model also reveals a counter-cyclical distribution margin, consistent with the benchmark model and the observed data, and shows that much of this counter-cyclical behavior is attributable to service-sector productivity shocks.

We further demonstrate the quantitative significance of incorporating state-dependent distribution friction in understanding monetary policy shock transmission. By comparing the dynamic gains from reducing distribution friction after a money supply shock between models with state-dependent and time-invariant friction, we assess their respective impacts on impulse responses. Using counterfactual analysis, we quantify gains in impulse responses from reducing distribution friction following a positive real-money supply shock. Notably, these gains are much larger when distribution friction is state-dependent. This is intuitive, as changes in state-dependent friction trigger significant adjustments in marginal cost and the optimal reset price, leading to further responses in labor, output, and consumption. Overall, this analysis underscores the importance of tying distribution friction to the state of the economy.

Our paper makes several significant contributions. First, we introduce state-dependent distribution friction into a standard dynamic multi-sector model and examine the cyclical behavior of the distribution margin. We find it counter-cyclical, primarily driven by productivity shocks in the service sector. Second, we employ novel microdata on Brazilian fuel prices to empirically verify this cyclicity, and cross-validate it using U.S. historical data on farm food products and Indian survey data on manufacturing firms. Third, we contribute to the debate on the real effects of money by quantifying the impact of state-dependent distribution friction. We show that the effects of an expansionary monetary policy shock are significantly dampened as state-dependent distribution friction increases, in contrast to models that assume time-invariant

friction. Lastly, we add to the literature on the real effects of conventional monetary policy. We demonstrate that state-dependent distribution-sector frictions can weaken the transmission of conventional monetary policies by significantly reducing the real effects of increases in the money supply. In a nutshell, our findings highlight a novel mechanism through which sectoral productivity shocks interact with state-dependent distribution frictions to affect the real effects of monetary policy.³

Our paper complements recent literature by highlighting that state-dependent distribution friction is crucial for understanding the real effects of monetary policy shocks. In contrast, models with constant distribution friction significantly overestimate the real effects of money. Consistent with existing studies, we introduce state-dependent distribution friction into models that explicitly address the distribution sector ([Burstein et al., 2003](#); [Corsetti and Dedola, 2005](#)). Our findings, showing that state-dependent distribution friction dampens the impact of expansionary monetary policy shocks, echo [Campa and Goldberg \(2010\)](#), who empirically document that such friction significantly restricts the transmission of import prices to final consumption prices of imported products.⁴

Departing from open-economy studies like [Burstein et al. \(2003\)](#) and [Corsetti and Dedola \(2005\)](#), our closed-economy model focuses on monetary policy's real effects on macroeconomic variables such as consumption and employment. [Van Leemput \(2021\)](#) notes that internal trade barriers, including domestic distribution costs, account for an average of 40% of total trade barriers in large emerging economies like India. This underscores the significance of internal distribution costs in a closed economy.⁵

Our paper further contributes to the extensive literature on the quantitative importance of product-market frictions for business-cycle movements. Labor market distortions, such as wage stickiness and matching frictions, have become increasingly recognized as important

³Along this dimension, our work also complements the related literature documenting the interplay between sectoral reallocation and real business cycle dynamics (see, for example, [Vu and Wu, 2020](#) and [Chodorow-Reich and Wieland, 2020](#)).

⁴Broadly, our findings align with the extensive literature emphasizing the importance of local distribution costs in understanding incomplete exchange rate pass-through ([Corsetti and Dedola, 2005](#); [Choudhri et al., 2005](#); [Campa and Goldberg, 2010](#)).

⁵For a comprehensive review on the significance of internal trade costs, see [Peter and Ruane \(2022\)](#).

factors in explaining labor market dynamics and macroeconomic issues.⁶ Recently, [Bils et al. \(2018\)](#) highlighted that product-market distortions, like price markups, are significant drivers of business cycles during recessions. Building on conventional price stickiness ([Bils and Klenow, 2004](#)) and price markups ([Bils et al., 2018](#)), our paper introduces a new source of product-market distortion: state-dependent distribution friction.⁷

Our work is closely related to the literature on variable markups, which emphasizes that observed price dynamics reflect not only marginal costs but also endogenous movements in markups driven by demand conditions and market structure. Studies such as [Arkolakis et al. \(2019\)](#) and [Atkeson and Burstein \(2008\)](#) show how markups vary with demand elasticities and strategic pricing. From this perspective, our state-dependent distribution friction can be interpreted as an effective contributor to variable retail markup, consistent with this literature.

Moreover, our contribution highlights a distinct mechanism for markup variation following [Corsetti and Dedola \(2005\)](#). Rather than relying on non-CES demand or firm-level pricing power, we show that state-dependent distribution frictions *per se* can also generate variable markups even when producer pricing is standard. As formalized in our model below, fluctuations in distribution frictions transmit macroeconomic shocks into consumer prices through the distribution sector. This mechanism complements existing variable-markup models by emphasizing distribution frictions as an independent and empirically relevant source of markup variation, with direct implications for the real effects of monetary policy.⁸

Our paper is also related to the broad, growing literature on the intersection between distribution margins and international trade ([Atkeson and Burstein, 2008](#); [Burstein et al., 2003](#);

⁶See, for example, [Galí et al. \(2007\)](#) on the welfare costs of business cycle fluctuations, [Pissarides \(2009\)](#) on the unemployment volatility puzzle, and [Karabarbounis \(2014\)](#) on international business cycle co-movements.

⁷[Crucini et al. \(2010\)](#) explicitly model both nominal rigidities and transportation costs to examine their roles in the variation of LOP (law of one price) deviations. However, unlike our state-dependent distribution friction, they assume a time-invariant transportation cost, modeled as the traditional iceberg cost.

⁸On the empirical side, our work is also related to studies documenting that markups vary systematically with preferences, local market conditions, and distribution frictions, generating incomplete exchange rate pass-through and persistent deviations from the law of one price (LOP). For example, [Alessandria and Kaboski \(2011\)](#) document higher markups in markets with higher income and wages, helping explain deviations from the LOP and absolute PPP. [Yilmazkuday \(2016\)](#) shows that cross-location price dispersion is driven largely by variable markups rather than marginal cost differences, while [Yilmazkuday \(2017\)](#) and [Yilmazkuday \(2019\)](#) demonstrate that endogenous markup adjustment helps explain weak price responses to globalization and international elasticity puzzles. Our framework contributes to this literature by modeling distribution frictions as a structural source of variable markups ([Nekarda and Ramey, 2013](#); [Gilchrist et al., 2017](#)).

Corsetti and Dedola, 2005; Berman et al., 2012; Cavallo et al., 2025) and to the spatial dimension of distribution margins (Redding, 2025; Alfaro and Chor, 2025). In a recent notable contribution, Cuba-Borda et al. (2025) similarly discuss the impact of trade costs for inflation dynamics in international trade models, where they recover what they call trade costs from bilateral trade flows, as is common in international trade, but they do not provide direct evidence on distribution costs.

The rest of the paper is organized as follows. Section 2 presents the benchmark dynamic multi-sector model with state-dependent distribution friction and establishes the countercyclicity of the distribution margin. Section 3 employs micro-price data from Brazil to test the model's theoretical prediction regarding the cyclicity of the distribution margin, and corroborates the findings with historical data on U.S. farm food products and survey data from Indian manufacturing firms. Sections 4 and 5 augment the benchmark model with price and wage rigidities to assess the quantitative importance of state-dependent distribution friction for the transmission of monetary policy shocks. Section 6 concludes.

2. A Multi-sector Model with State-dependent Distribution Friction

We consider a dynamic multi-sector model featuring state-dependent distribution friction, where the distribution sector uses differentiated services as intermediate inputs to deliver manufactured goods from producers to retailers. A key prediction of the benchmark model is that the distribution margin is counter-cyclical, a result we empirically verify in Section 3.

2.1. Production

The economy consists of two production sectors: manufacturing (m) and services (s). Each sector comprises a continuum of firms over the measure $[0,1]$, producing and selling either manufactured goods or services under monopolistic competition (Dixit and Stiglitz, 1977). Labor is the sole input, represented as $L_t(i, j)$, where sector i can be either manufacturing (m) or services (s), and index j denotes the type of labor input demanded by the firm. The production function for both manufactured goods and services is specified in the form of a constant elasticity

of substitution (CES):

$$Y_t^m = Z_{M,t} \left[\int_0^1 L_t(m, j)^{\frac{\phi-1}{\phi}} dj \right]^{\frac{\phi}{\phi-1}}, \quad Y_t^s = Z_{S,t} \left[\int_0^1 L_t(s, j)^{\frac{\phi-1}{\phi}} dj \right]^{\frac{\phi}{\phi-1}}, \quad (1)$$

where ϕ is the elasticity of substitution across different types of labor. In Eqn. (1), $Z_{M,t}$ and $Z_{S,t}$ denote the productivity levels in the manufacturing and service sectors, respectively.

We model manufactured goods and services separately to reflect their distinct relationships with the distribution sector. Manufacturing faces significant distribution friction because producers and retailers are often spatially separated. In contrast, services do not require a distribution sector since the producers and retailers are the same. In this paper, the delivery business of the distribution sector is treated as a type of aggregated service, consistent with [Burstein et al. \(2003\)](#) and [Corsetti and Dedola \(2005\)](#). We introduce distribution friction by assuming that η_t units of distribution services in physical quantities are required to deliver one physical unit of manufactured goods from producers to retailers, and we conceptualize the proportion η_t through the following CES aggregation over differentiated services:

$$\eta_t = \left[\int_0^1 \eta_t(s)^{\frac{\theta-1}{\theta}} ds \right]^{\frac{\theta}{\theta-1}}, \quad (2)$$

where θ represents the elasticity of substitution across various differentiated services required to distribute manufactured goods.⁹ The proportional assumption aligns with [Burstein et al. \(2003\)](#) and [Corsetti and Dedola \(2005\)](#), following [Tirole \(1988\)](#), who noted that production and distribution are complements, typically consumed in fixed proportions. This means that the distribution service exhibits Leontief technology, reflecting the complementarity between production and distribution. Departing from [Burstein et al. \(2003\)](#) and [Corsetti and Dedola \(2005\)](#), who assume time-invariant distribution friction ($\eta_t = \eta$), we intentionally use the subscript t for η_t to indicate its dependency on the state of the economy.

⁹To name a few, real-world examples of such differentiated services include transportation, warehousing and inventory management, logistics coordination, as well as financial and risk management.

2.2. Household

The household engages in consumption, maintains real balances, and participates in the labor market. We denote the type of labor supplied by a representative household as j . Household j maximizes its lifetime expected utility in the following manner:

$$\mathcal{U}_t(j) = E_t \beta^{k-t} \sum_{k=t}^{\infty} \left[\ln C_k(j) + \chi \ln \left(\frac{M_k(j)}{P_k} \right) - \kappa L_k(j) \right], \quad (3)$$

where β represents the subjective discount factor, and $C_t(j)$ denotes the consumption composite as defined in Eqn. (5). We include real balances $M_t(j)/P_t$ to characterize the role of money in utility. Parameters χ and κ govern the respective importance of utility derived from holding real balances and the disutility associated with labor.

The household consumes a bundle of all manufactured goods and services. This aggregate consumption, denoted as $C_t(j)$, is organized within a nested structure. Specifically, $C_t(m, j)$ represents the consumption of manufactured good m , and $C_t(s, j)$ represents the consumption of service s by household j at time t . The total consumption of manufactured goods, $C_{M,t}(j)$, and services, $C_{S,t}(j)$, follows a standard CES aggregation with an identical elasticity of substitution, θ , as employed in the distribution sector:

$$C_{M,t}(j) = \left[\int_0^1 C_t(m, j)^{\frac{\theta-1}{\theta}} dm \right]^{\frac{\theta}{\theta-1}}, \quad C_{S,t}(j) = \left[\int_0^1 C_t(s, j)^{\frac{\theta-1}{\theta}} ds \right]^{\frac{\theta}{\theta-1}}. \quad (4)$$

The overall consumption bundle $C_t(j)$ is a Cobb-Douglas aggregation of manufacturing and service consumption, where parameter γ dictates the expenditure proportion allocated to manufactured goods:

$$C_t(j) = \frac{C_{M,t}(j)^\gamma C_{S,t}(j)^{1-\gamma}}{\gamma^\gamma (1-\gamma)^{1-\gamma}}. \quad (5)$$

Let the prices of manufactured goods m and services s at time t be denoted as $p_t(m)$ and $p_t(s)$, respectively. Given the utility and consumption structures described earlier, the price

indices for manufactured goods ($P_{M,t}$) and services ($P_{S,t}$) can be expressed as:

$$P_{M,t} = \left[\int_0^1 [p_t(m)]^{1-\theta} dm \right]^{\frac{1}{1-\theta}}, \quad P_{S,t} = \left[\int_0^1 [p_t(s)]^{1-\theta} ds \right]^{\frac{1}{1-\theta}}. \quad (6)$$

The overall price index (P_t) constitutes a Cobb-Douglas aggregation of the price indices for manufactured goods ($P_{M,t}$) and services ($P_{S,t}$):

$$P_t = P_{M,t}^\gamma P_{S,t}^{1-\gamma}. \quad (7)$$

Household j maximizes the objective function $\mathcal{U}_t(j)$ under a budget constraint. To fund consumption, the household gains revenue from holding bonds $(1 + i_t)B_t(j)$ and from labor $W_t(j)L_t(j)$. Additionally, the household earns profits from firms but must also cover a lump-sum tax $T_t(j)$. The budget constraint for household j can be expressed as:

$$P_{M,t}C_{M,t}(j) + P_{S,t}C_{S,t}(j) + M_t(j) + B_t(j) = M_{t-1}(j) + (1 + i_t)B_{t-1}(j) + W_t(j)L_t(j) + \int_0^1 \Pi_t(m, j)dm + \int_0^1 \Pi_t(s, j)ds - T_t(j), \quad (8)$$

where $\int_0^1 \Pi_t(m, j)dm + \int_0^1 \Pi_t(s, j)ds$ denotes the profits from the two production sectors. Notably, the distribution sector operates under perfect competition, resulting in zero profits for the household from this sector.¹⁰

2.3. Government

We assume away government spending. Seigniorage revenue is rebated to the household through the lump-sum tax $T_t(j)$. Consequently, the lump-sum tax is negative when seigniorage

¹⁰Burstein et al. (2003) and Corsetti and Dedola (2005) likewise propose a competitive distribution sector. Furthermore, De Loecker and Scott (2016) analyze the markup imposed by various supply-side entities (producers, distributors, and retailers) in the U.S. brewing industry. They conclude that the distribution markup is negligible, constituting less than 8% of the total markup.

revenue is positive. The government's budget constraint is specified as:

$$\int_0^1 [M_t(j) - M_{t-1}(j)]dj + \int_0^1 T_t(j)dj = 0. \quad (9)$$

For simplicity, we denote the total consumption expenditure as $\mu_t(j) \equiv P_t C_t(j)$. Monetary policy is embedded and characterized within the optimal conditions of the representative household j as follows:

$$\frac{1}{\mu_t(j)} = \beta(1 + i_{t+1})E_t\left(\frac{1}{\mu_{t+1}(j)}\right), \quad (10)$$

$$M_t(j) = \chi \frac{1 + i_{t+1}}{i_{t+1}} \mu_t(j), \quad (11)$$

where the former stems from the Euler equation governing intertemporal consumption choice, while the latter determines the optimal holding of nominal balances by the household.

2.4. Optimal Wages and Prices in A Flexible Wage and Price Model

In this benchmark model, wages are perfectly flexible, an assumption we will revisit and relax later in Section 4. The optimal wage condition for the representative household j is:

$$W_t(j) = \zeta \mu_t(j). \quad (12)$$

Given that the distribution of manufactured goods necessitates η_t units of aggregated services, the demand for services s by the representative household comprises the sum of the following two components:¹¹

$$C_t(s) + \eta_t = \left(\frac{p_t(s)}{P_{S,t}}\right)^{-\theta} \left(C_{S,t} + \eta_t \int_0^1 C_t(m)dm\right), \quad (13)$$

where the retail price, equivalent to the producer price, of service s is set as a constant markup

¹¹For simplicity, we will omit the notation j for the representative household going forward. Thus, $C_t(s)$ in Eqn. (13) corresponds to $C_t(s, j)$ in Eqn. (4).

over the unit labor cost:

$$p_t(s) = P_{S,t} = \frac{\theta}{\theta - 1} \frac{W_t}{Z_{S,t}}. \quad (14)$$

For manufacturing firms, considering our assumption of a perfectly competitive distribution sector, the retail price of a manufactured good m can be represented as:

$$p_t(m) = \bar{p}_t(m) + \eta_t P_{S,t}, \quad (15)$$

where $\bar{p}_t(m)$ denotes the producer price of manufactured good m . The profit maximization problem faced by the firm producing a manufactured good m can be stated as:

$$\max_{\bar{p}(m)} \bar{p}_t(m) C_t(m) - \frac{W_t}{Z_{M,t}} C_t(m),$$

where

$$C_t(m) = \left(\frac{\bar{p}_t(m) + \eta_t P_{S,t}}{P_{M,t}} \right)^{-\theta} C_{M,t}. \quad (16)$$

Solving the profit maximization problem, we derive the optimal producer price of the manufactured good m as:

$$\bar{p}_t(m) = \frac{\theta}{\theta - 1} \left(1 + \frac{\eta_t}{\theta - 1} \frac{Z_{M,t}}{Z_{S,t}} \right) \frac{W_t}{Z_{M,t}}. \quad (17)$$

The retail price of a manufactured good m is determined by adding the distribution cost to the producer price, expressed as follows:

$$p_t(m) = \underbrace{\frac{\theta}{\theta - 1} \left(1 + \frac{\eta_t}{\theta - 1} \frac{Z_{M,t}}{Z_{S,t}} \right) \frac{W_t}{Z_{M,t}}}_{\text{Optimal Producer Price}} + \underbrace{\eta_t \frac{\theta}{\theta - 1} \frac{W_t}{Z_{S,t}}}_{\text{Distribution Cost}}. \quad (18)$$

To align the model with the empirical analysis in Section 3 and existing literature (e.g., [Burstein et al., 2003](#)), we define the distribution margin δ_t as the proportion of distribution cost

in the retail price of manufactured goods:

$$\delta_t = \frac{\eta_t P_{S,t}}{p_t(m)}. \quad (19)$$

Eqn. (19) encompasses three distinct variables on the distribution sector: distribution friction, distribution cost, and distribution margin. While these variables are closely interconnected, they exhibit some notable distinctions. η_t denotes the distribution friction, representing the real-term or physical quantity of aggregated services required to distribute one physical unit of manufactured goods. When multiplied by the price of services $P_{S,t}$, we obtain the distribution cost $\eta_t P_{S,t}$, which signifies the nominal cost of distributing one physical unit of manufactured goods. Lastly, the distribution margin is the ratio of distribution cost to the retail price of manufactured goods $p_t(m)$, representing the proportion of the retail price attributed to the distribution sector.

2.5. Equilibrium Conditions and Analytical Characterization

The equilibrium is characterized by a collection of variables $\{C_t, W_t, L_t, i_t, l_t, M_t, \mu_t, P_t, P_{M,t}, P_{S,t}, C_{M,t}, C_{S,t}, \Pi_t, T_t, B_t, \delta_t\}$ satisfying the following conditions: (i) the representative household maximizes utility; (ii) firms maximize profits; (iii) the government's budget constraint is met; and (iv) all markets clear. In this benchmark model, we summarize the cyclicity of the distribution margin with the following proposition. The detailed numerical analysis is deferred to Section 4, which incorporates price and wage rigidities.

Proposition 1 (*Countercyclical*). *In equilibrium, the distribution margin (δ_t) is counter-cyclical for variation driven by the distribution friction.*

To understand why **Proposition 1** holds, first notice that we have $Y_t = C_t$ in equilibrium, where the following condition determines aggregate consumption:

$$C_t = \zeta \frac{\theta - 1}{\theta} Z_{S,t} \left(\frac{Z_{S,t}}{Z_{M,t}} + \frac{\eta_t \theta}{\theta - 1} \right)^{-\gamma}. \quad (20)$$

Hence, $\frac{\partial Y_t}{\partial \eta_t} = \frac{\partial C_t}{\partial \eta_t} < 0$. Since, by definition, $\frac{\partial \delta_t}{\partial \eta_t} > 0$, it follows that $\frac{\partial \delta_t}{\partial Y_t} = \frac{\partial \delta_t}{\partial \eta_t} \times \frac{\partial \eta_t}{\partial Y_t} < 0$. This result establishes the countercyclicality of the distribution margin in equilibrium. Inspired by **Proposition 1**, we posit that the distribution friction varies with the state of the economy (i.e., $\eta_t \equiv \eta(\Omega_t)$), where Ω_t represents the set of state variables at time t . To ensure tractability, we also assume that the distribution friction function is increasing and concave.¹²

3. The Counter-cyclical Distribution Margin: Evidence from Microdata

In this section, we use microdata from Brazil to examine the benchmark model's prediction that the distribution margin is counter-cyclical. Our dataset provides weekly, state-level information on the various components of Brazilian fuel prices. We further corroborate the benchmark empirical findings using annual historical data on farm food products from the United States and large-scale annual manufacturing survey data from India, offering cross-country, cross-period, and cross-sector validation of the empirical patterns.

3.1. Data and Summary Statistics

We combine weekly state-level Brazilian fuel price data from the *Fuel Price Survey* conducted by the *National Agency of Petroleum, Natural Gas and Biofuels* (ANP) with several state and national economic indicators from the *Brazilian Institute of Geography and Statistics*. The ANP gathers information from individual gas stations nationwide and issues weekly reports on gasoline, diesel, ethanol, and other fuel prices. This dataset provides detailed data on retail prices, distribution costs, gas station coverage, and fuel types. Our sample covers May 9th, 2004 to June 23rd, 2019. State-level price margins (including retail prices and distribution costs) are derived as averages across surveyed gas stations.¹³ In other words, each fuel price variable in our sample is measured, for a given state and week, as the average price across all gas stations operating within that state during that week. To analyze the cyclicity of the distribution margin, we

¹²We provide and discuss a detailed specification of the distribution friction function in Section 4.1.

¹³The survey is carried out through visits to the stations and the data collector must often take a photo of the price board (painel de preços) to audit the data. The collection of such data generally occurs in the first three business days of the week, with publication on Fridays. Source: <https://www.gov.br/anp/pt-br/assuntos/precos-e-defesa-da-concorrenca/precos/precos-revenda-e-de-distribuicao-combustiveis>. Last accessed Jan. 2025.

match fuel price data with macroeconomic indicators at varying frequencies (monthly, quarterly, yearly) and administrative levels (state, national).

In addition to being the most comprehensive and high-frequency microdata on distribution margins available, we use Brazilian fuel price data for several reasons. First, Brazil’s fuel sector accounts for approximately 11% of GDP and is a major source of energy for various industries. Second, as an emerging economy, Brazil relies heavily on fuels, with approximately 54% of energy consumption derived from gasoline, diesel, and ethanol. Third, fuel is a typical manufactured good that requires distribution due to geographical disparities between production and consumption. Fourth, fuel prices in Brazil are *de facto* regulated,¹⁴ allowing little room for price markup during distribution. Hence, the difference in prices charged by retailers and producers primarily reflects the actual unit distribution cost.¹⁵ Lastly, our dataset offers broad coverage across all 27 Brazilian states, implying that insights drawn from fuel price microdata hold significant implications for the national economy.¹⁶

While fuel prices directly contribute to distribution costs, we anticipate no significant bias from using fuel data that might affect our primary findings below. One rationale is the weak correlation (0.06) between fuel prices and distribution costs in the dataset. Furthermore, distribution entails additional key elements beyond fuel expenses, including equipment rental, depreciation, warehouse or storage facility costs, and local labor costs.

Following [Burstein et al. \(2003\)](#) and Eqn. (19), the distribution margin from the data is computed as the ratio of distribution cost to retail price:

$$DM_{s,f,t} = \frac{DC_{s,f,t}}{RP_{s,f,t}}, \quad (21)$$

where $DM_{s,f,t}$, $DC_{s,f,t}$, and $RP_{s,f,t}$ are the weekly distribution margin, distribution cost, and

¹⁴Although fuel prices were *de jure* deregulated in Brazil in 2002, the government still influences prices through Petrobras, a state-owned fuel producer and distributor. Specifically, the Central Bank of Brazil (BCB) continues to categorize fuel prices as regulated. For further details, see Banco Central do Brasil. Source: <https://www.bcb.gov.br/conteudo/home-en/FAQs/FAQ%2005-Regulated%20Prices.pdf>.

¹⁵Employing structural analysis, [De Loecker and Scott \(2016\)](#) also find that distributors in the U.S. brewing industry impose a markup that is relatively minor compared to those levied by producers and retailers.

¹⁶Additional details on data coverage for these 27 states are provided in Appendix Table A.1. The table reports state-level summary statistics for the distribution margin across all fuel products, including means, standard deviations, minimums, maximums, and the number of observations.

retail price of fuel type f in state s at week t .¹⁷ The distribution cost $DC_{s,f,t}$ is the difference between the retail price and the price at the factory gate (i.e., the producer price). We present a cross-sectional histogram and a time-series plot of the distribution margin in Fig. 1a and Fig. 1b, respectively. The two graphs indicate that the average distribution margins for gasoline, diesel, and ethanol typically range from 10% to 20% across both cross-sectional and temporal dimensions. Furthermore, we observe substantial variations in the distribution margin across states and over time for all fuel types.¹⁸

[Figure 1 here]

Consistent with Figures 1a and 1b, we observe that the average distribution margin in our sample is 16%. This aligns closely with the average reported by Campa and Goldberg (2010) for a selection of countries in the European Union, Australia, New Zealand, and the United States, based on aggregate data, falling within their documented range of 8% to 24%.¹⁹ In contrast, Burstein et al. (2003) find a substantially larger aggregate distribution margin for Argentina and the United States, with the distribution margin for all goods in the United States reported at around 46%. Similarly, Berger et al. (2012) document U.S. distribution wedges ranging from 50% to 70%. Furthermore, Crucini and Landry (2019) report a similar figure for the U.S. aggregate consumption basket, with the median distribution margin across goods and services standing at 41%.²⁰

¹⁷Within our dataset, we categorize fuel into four types: hydrated ethanol, common gasoline, diesel, and other varieties.

¹⁸Moreover, Fig. A.1 in the accompanying Appendix illustrates considerable geographical diversity within our dataset. Typically, landlocked states experience greater distribution friction, likely reflecting that most refineries in Brazil are located near ports. This geographical variation diminished between 2004 and 2019, potentially attributable to infrastructure improvements that have substantially reduced transportation costs over the past two decades.

¹⁹Hummels (2001) examines sector-level average freight rates across a selection of Latin American countries, New Zealand, and the United States. He finds that, for the petroleum industry, the average freight rate (defined as the ratio of freight cost to import price) ranges from 9% to 22%. Notably, the average petroleum freight rate for Brazil is 16.9%, which closely matches the average distribution margin in our study.

²⁰Intriguingly, Crucini and Landry (2019) reports a similar magnitude for fuel using U.S. data, with the distribution margin for gasoline documented as 19%. Our estimate based on modern Brazilian data is also close in magnitude to that reported by Cheng et al. (2025), which is derived from historical data on the automobile industry in the early twentieth century.

3.2. Cyclicity of the Distribution Margin: Regression Specification

To assess the cyclicity of the distribution margin using Brazilian micro price data, we conduct the following regression:

$$g_{s,f,t}^{DM} = \beta_0 + \beta_1 g_{s,t}^{GDP} + \beta_2 g_{s,f,t-1}^{DM} + \mathbf{g}_{s,t}^X \boldsymbol{\theta} + D_{s,f,t} + v_{s,f,t}, \quad (22)$$

where $g_{s,f,t}^{DM}$ represents the weekly growth rate of the distribution margin for fuel prices in state s , week t , and fuel type f ; $g_{s,t}^{GDP}$ denotes the annual growth rate of GDP in state s . β_1 is the coefficient of interest. We include the lagged variable $g_{s,f,t-1}^{DM}$ to capture the persistence of distribution margin growth.

In Eqn. (22), $\mathbf{g}_{s,t}^X$ is a vector containing the growth rates of several control variables: national monthly industrial production, national monthly consumer price index, state-level monthly construction cost index, national monthly consumer confidence index, national monthly exchange rate (real effective exchange rate index), weekly WTI crude oil spot price, and weekly stock market index. Table 1 provides the summary statistics for the variables included in our benchmark regression.

[Table 1 here]

$D_{s,f,t}$ contains dummy variables for the following fixed effects: fuel type, state, and week. The week fixed effects account for seasonality, while the state fixed effects capture unobserved state-level heterogeneity. Since we pool all fuel types in the regressions, it is essential to include fuel-type fixed effects.

3.3. Cyclicity of the Distribution Margin: Estimation Results

The estimation results for Eqn. (22) are presented in Table 2. In these regressions, we pool all fuel types (gasoline, diesel, and ethanol) and include week, state, and fuel-type fixed effects. Standard errors are clustered at the state level.

Table 2 shows that the distribution margin is negatively correlated with state-level GDP

growth, with the coefficient being statistically significant at the 1% level. Specifically, a one-percent increase in annual state-level GDP is associated with a 1.2% annualized decrease in the distribution margin. This empirical evidence supports the benchmark model prediction in **Proposition 1** that the distribution margin is counter-cyclical.

[Table 2 here]

Table 2 also shows evidence of self-adjustment in the distribution margin. Specifically, higher growth in the distribution margin in the previous period predicts lower current growth. Additionally, there is a statistically significant negative correlation between inflation and the distribution margin, consistent with the comovements between output and price levels. Furthermore, the growth rate of construction costs is positively correlated with the distribution margin, reflecting that construction services are an input to distribution infrastructure; thus, an increase in construction costs raises distribution costs. Similarly, the growth rate of oil prices is positively associated with the distribution margin since oil contributes to the user cost of distribution equipment.

Table 2 further reveals a statistically significant negative correlation between the Brazilian real exchange rate index and the distribution margin. This suggests that an appreciation of the Brazilian *real* is associated with a reduction in the distribution margin. Intuitively, currency appreciation makes exports more costly, thereby reducing external demand for Brazilian manufactured goods and reducing pressure on the distribution sector. Finally, the table shows that the distribution margin is not significantly correlated with changes in consumer confidence or fluctuations in the Brazilian domestic stock market.

To test cyclicalities, our benchmark specifications in Table 2 use growth rates, which may capture dynamics beyond pure cyclical movements, potentially raising concerns that the results reflect spurious correlations - for example, that economically stronger states attract more fuel distributors, lowering distribution margins. To address this concern, we re-estimate the regressions using HP-filtered cyclical components for all variables.²¹ The results, reported in

²¹Given that the Hodrick-Prescott filter is not commonly used at the weekly frequency in the literature, we first aggregate our dataset to the monthly frequency and then apply this filter with a smoothing parameter of 14,400.

Table A.2, are highly consistent: the GDP coefficient remains negative, statistically significant, and comparable in magnitude to the benchmark estimates.

3.4. Additional Counter Cyclical Evidence from other Countries

In addition to the microdata from Brazil, we also leverage data from the U.S. and from India to cross-validate our findings. For brevity, we leave the exposition of such exercises in the appendix. Overall, we find that data evidence from both the U.S. and India to be generally consistent with our findings using Brazilian data in this section.

4. An Augmented Model with Price and Wage rigidities

This section extends the benchmark model described in Section 2 to incorporate price and wage rigidities. We estimate this augmented model using Brazilian macroeconomic data and then explore the cyclical behavior of the distribution margin through model-based simulations. Our first goal is to confirm that the augmented model can replicate the counter-cyclical behavior of the distribution margin - both a prediction of the benchmark model presented in Section 2 and a key empirical feature observed in the micro price data discussed in Section 3. Later, we will use the estimated quantitative model as a workhorse to examine the role of state-dependent distribution friction in the transmission of monetary policy in Section 5.

4.1. The Augmented Model

In the augmented model, we introduce price stickiness in the spirit of Calvo (1983). Following the approach of De Loecker and Scott (2016), we assume that the retailer is responsible for both producing and distributing manufactured goods. Each period, a retailer h producing and selling manufactured goods faces a probability φ of being unable to adjust its price. At the same time, it can change its price with a probability of $1 - \varphi$. This parameter φ represents the degree of price rigidities. In contrast, pricing for service firms remains flexible, as is standard in the literature. Therefore, the retailer's pricing problem for manufactured goods m can be written

as:

$$\max_{\bar{p}_t(m)} E_t \sum_{k=0}^{\infty} (\beta\varphi)^k \frac{U'(C_{t+k})}{U'(C_t)} \left[\frac{\bar{p}_t(m) + \eta_t P_{S,t}}{P_{M,t+k}} C_t(m, j) - mc_{t+k}(m) C_t(m, j) \right],$$

where $mc_t(m)$ represents the marginal cost and $C_t(m, j)$ denotes the demand for manufactured goods of brand m by the representative household j . The marginal cost for firm m can be expressed as:

$$mc_t(m) = \left(1 + \frac{\eta_t}{\theta - 1} \frac{Z_{M,t}}{Z_{S,t}} \right) \frac{W_t}{Z_{M,t}} + \eta_t \frac{W_t}{Z_{S,t}}, \quad (23)$$

and the demand for manufactured good m by the representative household j is:

$$C_t(m, j) = \left(\frac{\bar{p}_t(m) + \eta_t P_{S,t}}{P_{M,t}} \right)^{-\theta} C_{M,t}(j). \quad (24)$$

Following [Obstfeld and Rogoff \(2000\)](#) and [Corsetti and Dedola \(2005\)](#), we introduce wage rigidities by positing that workers and firms agree on contracts one period in advance. Consequently, the optimal wage condition for the household is:

$$W_t(j) = \frac{\phi}{\phi - 1} \frac{E_{t-1}[\kappa L_t(j)]}{E_{t-1}[L_t(j)/\mu_t(j)]}, \quad (25)$$

where ϕ represents the bargaining power of workers, while κ gauges workers' relative preferences between consumption and working.

To close the augmented model, we specify the evolution of manufacturing productivity, service productivity, and real money supply as follows:

$$\log(Z_{M,t}) = \rho_Z^M \log(Z_{M,t-1}) + \varepsilon_{M,t}, \quad (26)$$

$$\log(Z_{S,t}) = \rho_Z^S \log(Z_{S,t-1}) + \varepsilon_{S,t}, \quad (27)$$

$$\log(m_t) = (1 - \rho_m) \log(\bar{m}) + \rho_m \log(m_{t-1}) + \varepsilon_{m,t}, \quad (28)$$

where the real money supply is defined as $m_t = M_t/P_{t-1}$, with the steady-state level of money supply $\bar{m}$ determined by deep parameters of the model. For ease of solving the model, we normalize all endogenous variables by the aggregate price in the previous period, P_{t-1} . The stochastic shocks $\varepsilon_{M,t}$, $\varepsilon_{S,t}$, and $\varepsilon_{m,t}$ are assumed to be independent and identically distributed with a normal distribution. Solving the retailer's pricing problem entails recognizing that, when feasible, all retailers set the same price and that there are many of them. The remaining equilibrium conditions follow from the benchmark model with flexible prices and wages in Section 2.

Building upon **Proposition 1**, we specify the state-dependent distribution friction function as $\eta_t \equiv \eta(\Omega_t) = \bar{\eta} \times \left(1 + \xi \frac{P_{S,t}C_{S,t}}{P_{t-1}C_{t-1}}\right)^\iota$, where $\bar{\eta}$ governs the steady-state value of η_t , ξ determines the extent of state-dependency for the distribution friction, and ι represents the degree of concavity for the friction function.²² This formulation draws inspiration from the investment adjustment cost function specified in [Jermann and Quadrini \(2012\)](#).

Our specification of the distribution friction function is based on two underlying assumptions. First, the service sector is responsible for meeting the demands of both the distribution sector and households. Therefore, increased consumer demand for services increases distribution costs for manufactured goods. Normalizing the demand for services to aggregate demand ensures that the distribution friction is expressed in real, physical quantities rather than nominal terms, since it relies on the ratio of two nominal terms.²³ Second, the distribution sector benefits from economies of scale, indicated by the concave nature of the friction function. This implies that a larger distribution sector can lower unit delivery costs. This concept is consistent with the core assumption in [Antoniades and Zaniboni \(2016\)](#) that unit retail costs decline with scale and is further supported by recent findings from [Ganapati \(2025\)](#), which document clear evidence of scale economies in the U.S. wholesale sector.

We calibrate selected parameters and estimate the remaining parameters of the augmented model using quarterly Brazilian macroeconomic data from 1996:Q1 to 2020:Q1. Specifically, we

²² $\xi = 0$ corresponds to a scenario where the distribution friction remains constant over time, while $\xi > 0$ implies that the distribution friction varies with the state of the economy (i.e., Ω_t). To ensure the state-dependent friction function is concave, we assume $0 < \iota < 1$.

²³ In Section 5, we also explore an alternative specification of the distribution friction function by replacing the expenditure ratio with the consumption ratio. Our results remain robust.

set the elasticity of substitution across different types of labor, ϕ , to 3, following the literature (Behar, 2025; Havranek et al., 2024). Taking into account recent work by Kaplan and Schulhofer-Wohl (2018), we set the parameter governing the disutility associated with labor κ to be one. We calibrate the state-dependence parameter in the distribution friction ξ to 1.²⁴ The value for ι that governs the concavity of the distribution friction function is set to be 0.5 following Jermann and Quadrini (2012).²⁵

Apart from ϕ , κ , ξ , and ι , we estimate the rest of the parameters from the data. The raw data, sourced from St. Louis FRED, include the manufacturing sector’s growth rate, real GDP growth, and the inflation rate. All data are seasonally adjusted and aligned with their model counterparts.²⁶ In our preferred quantitative model, we estimate the key parameter governing the size of the state-dependent distribution friction (i.e., $\bar{\eta}$) and incorporate state-dependence for the distribution friction (i.e., $\xi = 1$). We compare this model to variants without state-dependency in the distribution friction, where we either calibrate $\bar{\eta}$ to achieve a steady-state distribution margin $\delta^{SS} = 0.16$ as observed in the Brazilian fuel price data or estimate it. As shown below, the preferred model is better aligned with the aggregate data from Brazil than the other variants, as indicated by the posterior model probabilities.

In selecting priors, we assign a mean of 0.99 to the subjective discount factor β , corresponding to an annualized interest rate of 4%. The Calvo (1983) parameter for price stickiness φ and the elasticity of substitution θ are assigned means of 0.67 and 5, respectively, aligning with prior studies on micro-price behaviors.²⁷ We choose the prior mean for $\bar{\eta}$ to be consistent with the average distribution margin in the Brazilian microdata. We adopt prior choices for χ and γ consistent with Corsetti and Dedola (2005). We use relatively non-informative priors for parameters governing exogenous stochastic processes, centered around 0.7 and assumed to be normally distributed for their persistence. Table 3 summarizes our chosen priors and the posterior estimates.

²⁴While $\xi = 1$ is a calibration choice, we explore how sensitive the model is to different choices of ξ in the Appendix.

²⁵Table A.3 summarizes the parameters that are calibrated in our benchmark augmented model.

²⁶The main results of our paper remain unchanged when we restrict the sample to 2004:Q2-2019:Q2, which is in line with the duration of our Brazilian micro-price data.

²⁷See, for example, Bils and Klenow (2004) and Hsieh and Klenow (2009).

[Table 3 here]

Building on the countercyclicality of the distribution margin predicted by the benchmark model in Section 2 and corroborated by Brazilian fuel price data in Section 3, our next step is to assess whether our augmented quantitative model with price and wage rigidities aligns with this phenomenon. However, before proceeding, we want to demonstrate that our preferred quantitative model, which estimates the size parameter $\bar{\eta}$ and incorporates state dependence in the distribution friction, fits the data better than alternative models. Table A.6 shows that this preferred model attains the highest posterior model probability among the variants considered, making it a strong workhorse for analyzing the real effects of monetary policy shocks.

In addition, we assess robustness with respect to alternative calibrations of the curvature parameter ι and the state-dependence parameter ξ . The benchmark estimates remain highly stable across these variations. Tables A.7 and A.8 indicate that adjusting ι upward or downward has a very limited effect on the estimated parameters, and Tables A.9 and A.10 confirm similar robustness when smaller or larger values of ξ are considered.

4.2. The Countercyclicality of the Distribution Margin

Fig. A.2 presents a scatter plot of the distribution margin and output growth rates generated by the preferred model using the estimated parameters in Table 3, with 10,000 iterations and all shocks included. The figure shows that the distribution margin is predicted to be counter-cyclical in the estimated model, as indicated by the negative slope coefficient (-0.35). This finding aligns with the negative relationship between the distribution margin and output predicted by the benchmark model in **Proposition 1** and observed in the micro-price data presented in Table 2.²⁸

To explore the sources of the countercyclicality of the distribution margin, we examine how selected macroeconomic aggregates respond to an increase in the distribution margin induced

²⁸Fig. A.2 also illustrates the importance of allowing distribution frictions to be state dependent. In the same figure, we report results from an alternative model in which distribution frictions are assumed to be time invariant. Re-estimating this model fails to generate a counter-cyclical distribution margin; instead, the distribution margin becomes pro-cyclical, with the slope coefficient shifting from -0.35 in the state-dependent specification to 0.95 under time-invariant distribution frictions. This comparison highlights the essential role of state dependence in matching the empirical countercyclicality observed in the data.

by changes in the productivity of either the manufacturing or the service sector. Fig. 2 shows the responses of selected variables to a 1% increase in manufacturing productivity (left panel) and a 1% decrease in service productivity (right panel).²⁹ The use of these contrasting shocks is intentional, as both result in positive responses of the distribution margin δ_t . All responses in Fig. 2 are expressed as percentage deviations from their respective steady-state values.

[Figure 2 here]

Examining the details, we find that a positive shock to manufacturing productivity increases both the distribution margin and output. This indicates that a manufacturing productivity shock can generate procyclicality in the distribution margin. In contrast, a negative shock to service productivity increases the distribution margin but decreases output, inducing countercyclicality in the distribution margin. Combined with the overall countercyclicality reported in Fig. A.2 in the Appendix, this suggests that service sector productivity shocks dominate the countercyclicality of the distribution margin in the estimated model.³⁰ This result is intuitive, given that the distribution sector relies on various differentiated services as intermediate inputs to deliver manufactured goods.

To further investigate the sources of the countercyclicality of the distribution margin, Fig. 3 decomposes the historical variations in output (Y_t) and the distribution margin (δ_t) into contributions from manufacturing productivity, service productivity, and money supply shocks. Three key insights stand out. First, output and the distribution margin are overall negatively correlated, aligning with the findings in Fig. A.2. Second, the majority of the variation in the distribution margin is driven by service productivity shocks. This model-based finding resonates with our direct empirical evidence in Table A.4, which shows that service-sector growth is the primary driver of the distribution margin's counter-cyclical behavior in Brazilian data. Third, only service productivity shocks generate opposite variations in output and the distribution margin. This further indicates that the countercyclicality of the distribution margin predicted

²⁹The impulse responses of other variables to these two shocks are presented in Fig. A.4.

³⁰We focus on these two productivity shocks due to the much smaller direct response of the distribution margin to a shock in money supply, which can also be inferred from Fig. 3 below.

by the model is largely due to changes in service sector productivity.³¹

[Figure 3 here]

In the Appendix, we use the Brazilian fuel data to directly test whether the service sector is primarily responsible for the countercyclicality of the distribution margin. To this end, we collect annual growth rates of manufacturing value added and services value added from the *Brazilian Institute of Geography and Statistics* (IBGE). Ideally, we would rely on state-level annual growth rates for both sectors, but such data do not exist; the highest-frequency, highest-quality series available are aggregate annual measures for the national economy. Table A.4 reports the results. The estimates clearly show that service-sector growth, rather than manufacturing growth, is the primary driver of the distribution margin's counter-cyclical behavior.

5. Monetary Policy under State-dependent Distribution Friction

We now examine the quantitative importance of state-dependent distribution friction for the real effects of money using the preferred quantitative model from Section 4 as a workhorse. To do this, we explore three scenarios for steady-state distribution friction: $\eta^{ss} = 0, 0.13$, and 0.60 . These correspond to steady-state distribution margins (δ^{ss}) of $0, 0.16$, and 0.46 , respectively. The first scenario represents the absence of distribution friction, while the second scenario ($\delta^{ss} = 0.16$) reflects the average distribution margin observed in Brazilian micro-price data. The third scenario ($\delta^{ss} = 0.46$) matches the average distribution margin in U.S. aggregate data as reported by [Burstein et al. \(2003\)](#).

In Fig. 4, we illustrate the responses of manufacturing output, aggregate consumption, the distribution margin, labor, the price of manufactured goods, real wages, and inflation to a 10% positive shock to the real money supply ($\varepsilon_{m,t}$). All impulse responses are expressed as percentage deviations from their respective steady states except for inflation.

[Figure 4 here]

³¹On a side note, these patterns also imply the importance of state-dependent distribution friction for understanding the effects of productivity shocks in more general settings - a topic we leave for future research.

A key insight from Fig. 4 is that the effects of a positive money supply shock are dampened as η^{ss} increases. Specifically, the responses of all key variables are most pronounced when distribution friction is absent (solid line). These responses become less significant as η^{ss} increases (i.e., $\eta^{ss} = 0.13$ and 0.60). This pattern suggests that the real impacts of money decreases as the size of the distribution friction increases in our preferred quantitative model.

Since the three cases differ only in the magnitude of the steady-state distribution friction, this result demonstrates that ignoring or underestimating distribution friction leads to misleading effects of monetary policy shocks. Specifically, it results in artificially larger real effects of these shocks, thereby overestimating the transmission of monetary policies into the real economy.

In our preferred quantitative model, the extent to which distribution friction dampens the responses of output and other aggregate variables largely depends on the feedback mechanism between the manufacturing and service sectors, rather than the direct effect on the distribution margin itself. Fig. 4 illustrates that the response of the distribution margin is much weaker than the responses of manufacturing output, aggregate consumption, and labor.³² This can be understood by examining the following equilibrium condition that combines the budget constraint and the firm's pricing condition:

$$w_t L_t = \frac{mc_t^R(\eta_t)}{p_{M,t}}(p_{M,t} - \eta_t p_{S,t})C_{M,t} + \left(\frac{\theta - 1}{\theta}\right)p_{S,t}(C_{S,t} + \eta_t C_{M,t}), \quad (29)$$

where w_t denotes the real wage, $p_{M,t}$ and $p_{S,t}$ are real manufacturing and service prices normalized by the aggregate price level, and $mc_t^R(\eta_t)$ stands for the real marginal cost, which varies with η_t . Intuitively, Eqn. (29) implies that the distribution friction η_t might affect the marginal propensity to consume by altering marginal costs and consequently the optimal reset price. Moreover, because the distribution friction is assumed to be contingent on the state of the economy through the expenditure ratio, Eqn. (29) indicates a feedback loop between consumption and the state-dependent distribution friction.

When the money supply expands, the distribution friction initially declines, consistent with the distribution margin's general countercyclicality. However, the marginal effect of this de-

³²Notice again that this is consistent with Fig. 3, as we mentioned earlier.

crease in distribution friction on consumption becomes less pronounced as steady-state distribution friction increases. This interaction, coupled with the feedback loop implied in Eqn. (29), results in the restrained reactions of output, consumption, wage, and labor illustrated in Fig. 4.

Indeed, an analytical result concerning the mitigated impacts of increased distribution friction on consumption can be derived from the benchmark model introduced in Section 2. Assuming perfect foresight and the absence of nominal rigidities, we denote the marginal effects of distribution friction on consumption as $ME_t^{C,\eta} \equiv \frac{\partial C_t}{\partial \eta_t} < 0$.³³ We then derive:

$$\frac{\partial ME_t^{C,\eta}}{\partial \eta_t} = \gamma(\gamma + 1)\zeta \frac{\theta}{\theta - 1} Z_{S,t} \left(\frac{Z_{S,t}}{Z_{M,t}} + \frac{\eta_t \theta}{\theta - 1} \right)^{-\gamma-2} > 0,$$

which implies that the response of consumption to a real money supply shock is dampened as distribution friction increases. This result is evident from Fig. 4, where the response of aggregate consumption attenuates as η^{ss} increases. Consequently, this implies weaker responses in the demand for manufactured goods, labor, wages, and prices.³⁴ Broadly, the dampening effect of distribution friction aligns with the findings of [Campa and Goldberg \(2010\)](#), who show that distribution costs suppress the transmission of import prices into the retail prices of final goods.³⁵

While not central to our key finding on the dampening effect of a higher distribution friction, it is reassuring that the hump-shaped responses of aggregate variables in Fig. 4 align with the presence of price and wage rigidities in the model. Specifically, both output and consumption rise following a 10% increase in the real money supply, with responses peaking in the second quarter after the shock and then gradually returning to their steady states.

³³Given $C_t = \zeta \frac{\theta-1}{\theta} Z_{S,t} \left(\frac{Z_{S,t}}{Z_{M,t}} + \frac{\eta_t \theta}{\theta-1} \right)^{-\gamma}$, the marginal consumption is $\frac{\partial C_t}{\partial \eta_t} = -\gamma \zeta Z_{S,t} \left(\frac{Z_{S,t}}{Z_{M,t}} + \frac{\eta_t \theta}{\theta-1} \right)^{-\gamma-1} < 0$.

³⁴The main results remain consistent when the distribution friction function is specified alternatively based on the consumption ratio instead of the expenditure ratio. See Fig. A.5 for further details. Moreover, motivated by a suggestion from an anonymous referee, we further assess robustness by re-specifying the distribution friction function to depend exclusively on current expenditures and, alternatively, on productivity shocks in the service sector. Figures A.6 and A.7 present the corresponding results. The responses remain qualitatively unchanged under the current-expenditure specification and are largely preserved when conditioning on service-sector productivity shocks.

³⁵Furthermore, our findings are consistent with [Choudhri et al. \(2005\)](#), who demonstrate that distribution costs dampen the responses of wages and retail prices to exchange rate variations.

5.1. The Importance of State-dependency in Distribution Friction

We next investigate the quantitative importance of incorporating state-dependency into the distribution friction by conducting two counterfactual exercises. These exercises aim to illustrate the value added of introducing state-dependent distribution friction. The key difference between them is whether the distribution friction is allowed to depend on the state of the economy.

In the first exercise, we examine the model with state-dependent distribution friction where the size parameter (i.e., $\bar{\eta}$) in the specified distribution friction function is estimated, and the curvature and state-dependence parameters (i.e., ι and ξ) are calibrated as in the benchmark preferred quantitative model. We then calculate the relative gains in the impulse responses of selected variables following a 10% increase in the real money supply when we reduce the steady-state value of the distribution friction (η^{ss}) from 13% to 0%.³⁶ In the second exercise, we use the model with time-invariant distribution friction and compute the gains of endogenous variables when we similarly reduce the steady-state value of the distribution friction (η^{ss}) from 13% to 0% following the same 10% real money supply shock.³⁷ These two exercises allow us to assess the quantitative importance of introducing state-dependent distribution friction by explicitly controlling for the magnitude of the friction.

Specifically, we denote the impulse response of variable x_t at horizon h after the monetary shock as $\frac{\partial x_{t+h}}{\partial \varepsilon_{m,t}}$. The gain at horizon h when we decrease the steady-state distribution friction (η^{ss}) from 13% to 0% is defined as follows:

$$g_h(x) = \left(\frac{\partial x_{t+h}^{\eta^{ss}=0}}{\partial \varepsilon_{m,t}} - \frac{\partial x_{t+h}^{\eta^{ss}=0.13}}{\partial \varepsilon_{m,t}} \right) \bigg/ \frac{\partial x_{t+h}^{\eta^{ss}=0.13}}{\partial \varepsilon_{m,t}}. \quad (30)$$

We denote the gain at horizon h for the model with state-dependent distribution friction as $g_h^{SD}(x)$, and for the model with time-invariant friction as $g_h^{TI}(x)$. The difference in the gain at horizon h between the two models is then given by $g_h^{DID}(x) = g_h^{SD}(x) - g_h^{TI}(x)$.

³⁶Economically, this corresponds to moving from the average Brazilian distribution margin of $\delta^{ss} = 16\%$ to a frictionless economy with $\delta^{ss} = 0\%$.

³⁷In this case, we have $\xi = 0$ and $\iota = 0$. The only parameter that we preserve and estimate for the distribution friction is $\bar{\eta}$.

The dynamics of the difference in gains across the two models (i.e., $g_h^{DID}(x)$) at various horizons are presented in Table 4. The values are expressed as percentage gains relative to the benchmark distribution friction value (i.e., $\eta^{ss} = 0.13$).

[Table 4 here]

Table 4 demonstrates that the gains from a reduction in distribution friction are significantly larger when the model features state-dependent distribution friction, as indicated by positive values of $g_h^{DID}(x)$ for all selected endogenous variables at almost all horizons. Notice that a reduction in distribution friction within a model featuring time-invariant distribution friction can still propagate the effects of a money supply shock on labor, but it plays a negligible role in affecting other endogenous variables such as output, consumption, and prices. In contrast, the gains are substantial in the model with state-dependent distribution friction. Here, the reduction in state-dependent distribution friction leads to a significant change in the marginal cost, which then substantially affects prices through the price-setting conditions in equilibrium.³⁸

This substantial price effect subsequently triggers responses in real variables, such as labor, output, and consumption, through labor and goods demand functions. These responses, in turn, generate feedback into distribution friction via aggregate expenditures in the distribution friction function.³⁹ The propagation through the price channel, and therefore through consumption, serves as the primary mechanism through which the model with state-dependent distribution friction realizes larger gains compared to the model with time-invariant friction.⁴⁰ The contrast between these two counterfactual exercises clearly underscores the quantitative importance of incorporating state-dependency into the distribution friction.

³⁸The impact of state-dependent distribution friction on marginal cost is summarized by the equilibrium condition: $mc_t = \left(1 + \frac{\eta_t}{\theta - 1} \frac{Z_{M,t}}{Z_{S,t}}\right) \frac{W_t}{Z_{M,t}} + \eta_t \frac{W_t}{Z_{S,t}}$. This equation shows that any change in state-dependent distribution friction, η_t , can trigger a subsequent change in marginal cost.

³⁹The feedback to state-dependent distribution friction through aggregate expenditures is demonstrated by the distribution friction function: $\eta_t = \bar{\eta} \times \left(1 + \xi \frac{P_{S,t} C_{S,t}}{P_{t-1} C_{t-1}}\right)^l$. As it shows, an increase in total expenditures in the previous period ($P_{t-1} C_{t-1}$) can initiate a new-round reduction in distribution friction.

⁴⁰This price channel aligns with the findings in [Benedict et al. \(2020\)](#), which show that firms with higher distribution margins are less likely to reprice, thereby dampening the impacts of potential exogenous shocks.

5.2. Responses to Monetary Policy Shock under the Taylor Rule

We next examine the model's responses to an alternative monetary shock, applying the canonical Taylor rule across various values of the steady-state distribution friction η^{ss} . This serves as a robustness check against the use of money supply shocks in the baseline scenario. Specifically, we replace the exogenous money supply process with the following monetary policy rule:

$$\frac{1 + i_t}{1 + \bar{i}} = \left(\frac{1 + i_{t-1}}{1 + \bar{i}} \right)^{\rho_i} \left[\left(\frac{Y_t}{\bar{Y}} \right)^{\phi_y} \left(\frac{1 + \pi_t}{1 + \bar{\pi}} \right)^{\phi_\pi} \right]^{1-\rho_i} \varepsilon_t^T, \quad (31)$$

where the monetary shock ε_t^T is independently and identically distributed (i.i.d.) and follows a normal distribution. The parameter ρ_i determines the persistence of the monetary process, while ϕ_y and ϕ_π represent the weights on the output and inflation gaps, respectively. Fig. 5 shows the responses of selected variables to a 1% negative shock to the nominal interest rate. For this exercise, we set $\rho_i = 0.9$, $\phi_y = 0.5$, and $\phi_\pi = 1.5$, consistent with estimates from the literature (Taylor, 1993).

[Figure 5 here]

Similar to Fig. 4, Fig. 5 presents the responses of selected endogenous variables for three values of steady-state distribution friction: $\eta^{ss} = 0, 0.13, 0.60$, with the legends arranged accordingly. The responses are shown as percentage deviations from the steady states except for inflation. Fig. 5 demonstrates that higher distribution friction dampens the effects of an expansionary monetary policy shock following the Taylor rule. This result aligns with the responses in Fig. 4, where monetary policy is modeled as a money supply shock.

6. Conclusion

We study the quantitative importance of state-dependent distribution friction for the transmission of monetary policy shocks in a closed-economy version of the dynamic multi-sector model à la Corsetti and Dedola (2005), where the delivery of manufactured goods relies on differentiated intermediate inputs from the service sector. Departing from existing literature, we incorporate

state dependence into the distribution friction and find that the resulting distribution margin is counter-cyclical in our benchmark model.

We then use novel microdata on Brazilian fuel prices to show that the model's prediction of a counter-cyclical distribution margin is strongly supported by the data, highlighting an important empirical feature that has largely been overlooked in the existing literature, likely due to the absence of suitable micro-level price data. Estimating an augmented quantitative model with price and wage rigidities, we find that only the specification with state-dependent distribution frictions is able to replicate the observed countercyclicality, whereas an alternative model with time-invariant frictions fails to do so. We further show that productivity shocks originating in the service sector are the primary driver of this counter-cyclical behavior.

Employing the estimated preferred quantitative model as the workhorse, we then illustrate the quantitative importance of state-dependent distribution frictions for the transmission of monetary policy shocks, revealing its substantial dampening effects on macroeconomic variables in the real sector. Furthermore, we demonstrate, through counterfactual analyses, that state dependence in the distribution friction plays an important role in understanding the real effects of monetary policy shocks, even after explicitly controlling for the magnitude of the distribution friction. Overall, our findings highlight a novel mechanism through which sectoral productivity shocks interact with state-dependent distribution frictions in determining the real effects of monetary policy.

Data availability statement

The codes and data that support the findings of this study are available upon reasonable request.

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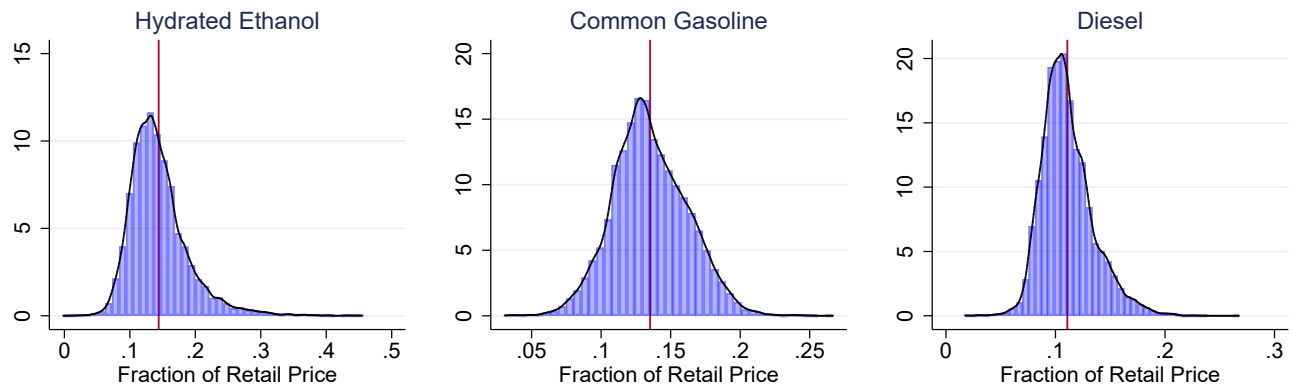
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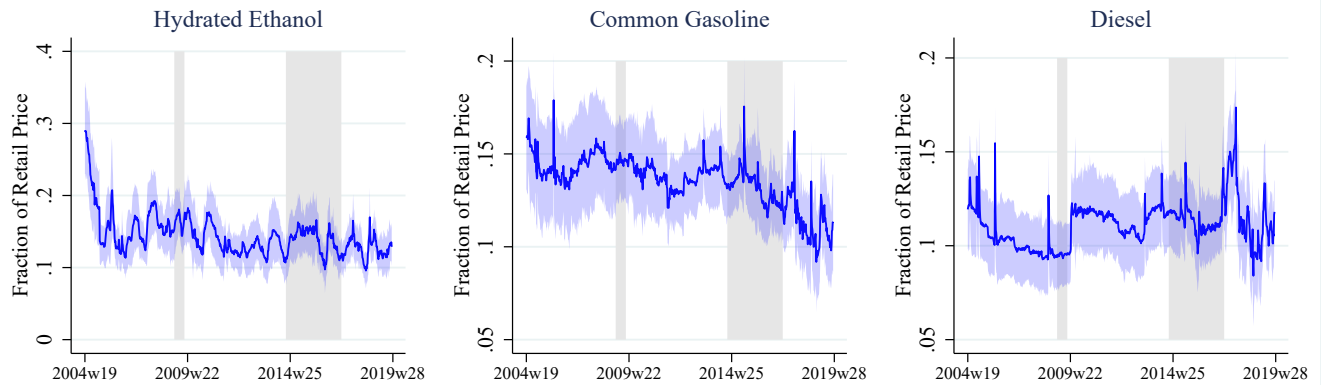
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Figure 1. The Distribution Margin across Locations and over Time

(a) The Distribution Margin across Locations

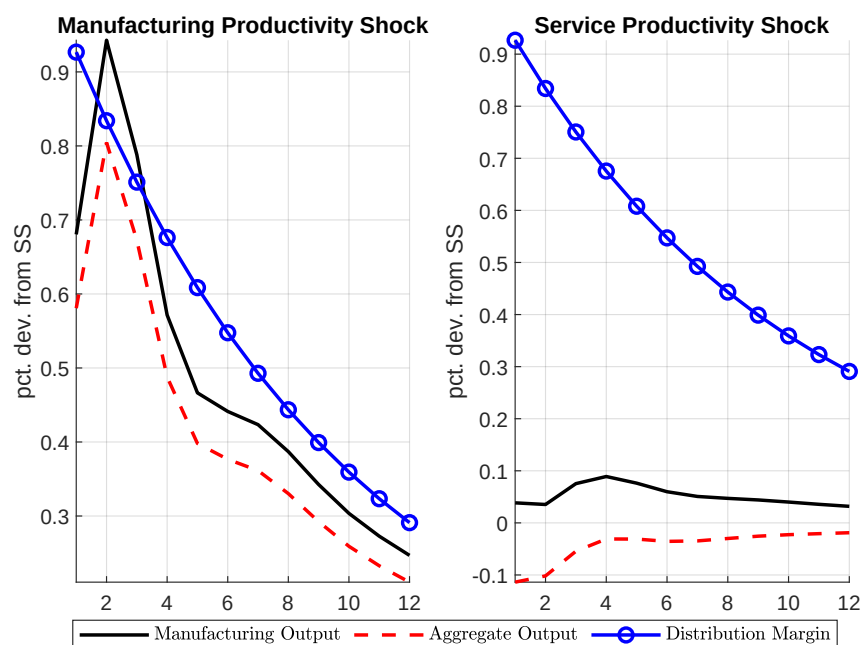


(b) The Distribution Margin over Time



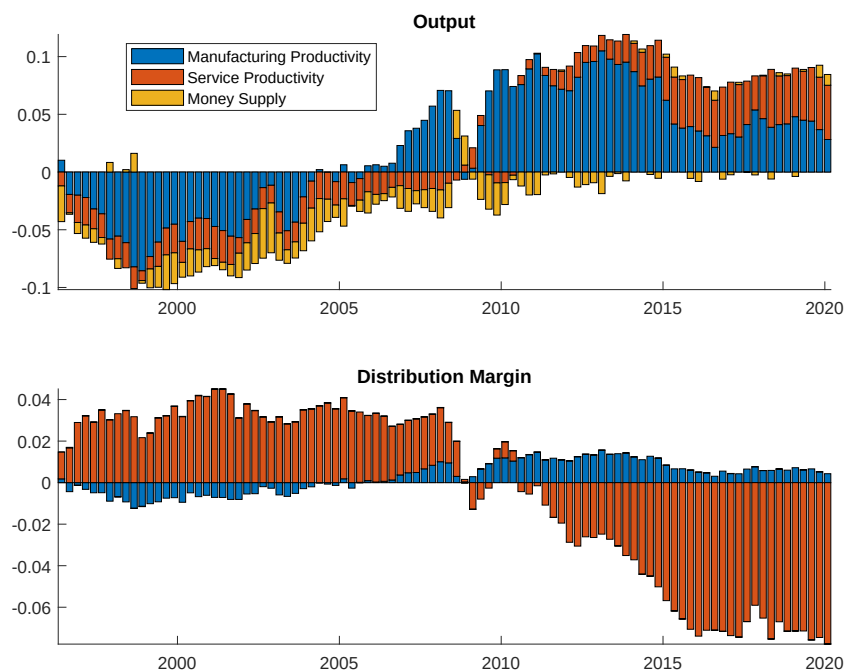
Note: The distribution margin represents the ratio of distribution cost to retail price. The upper panel shows the distribution of this margin across locations, while the lower panel shows its distribution over time. In Panel (b), the shaded areas denote the official Brazilian recessions as dated by the Brazilian economic cycle dating committee (CODACE) of the FGV IBRE.

Figure 2. Responses to Sector-specific Productivity Shocks



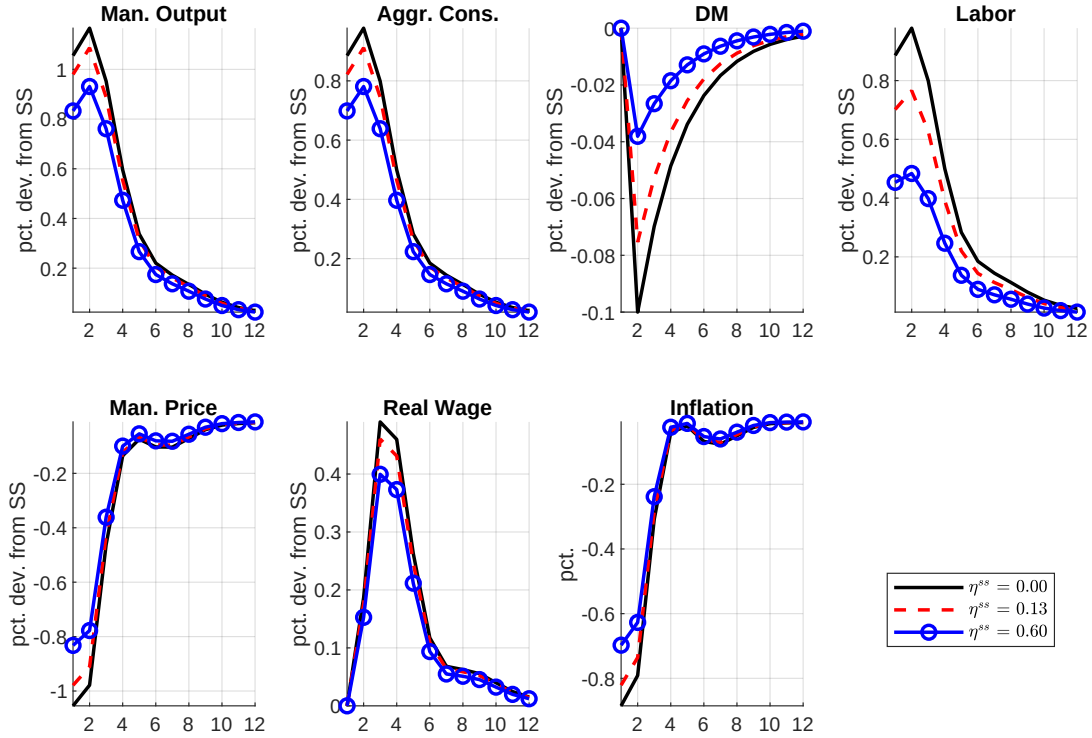
Note: Responses to a 1% increase in manufacturing productivity and a 1% decline in service productivity. The scale for all subfigures is expressed as a percentage deviation from the steady state. The horizontal axis measures the number of quarters following the shock.

Figure 3. Variance Decomposition: Output and the Distribution Margin



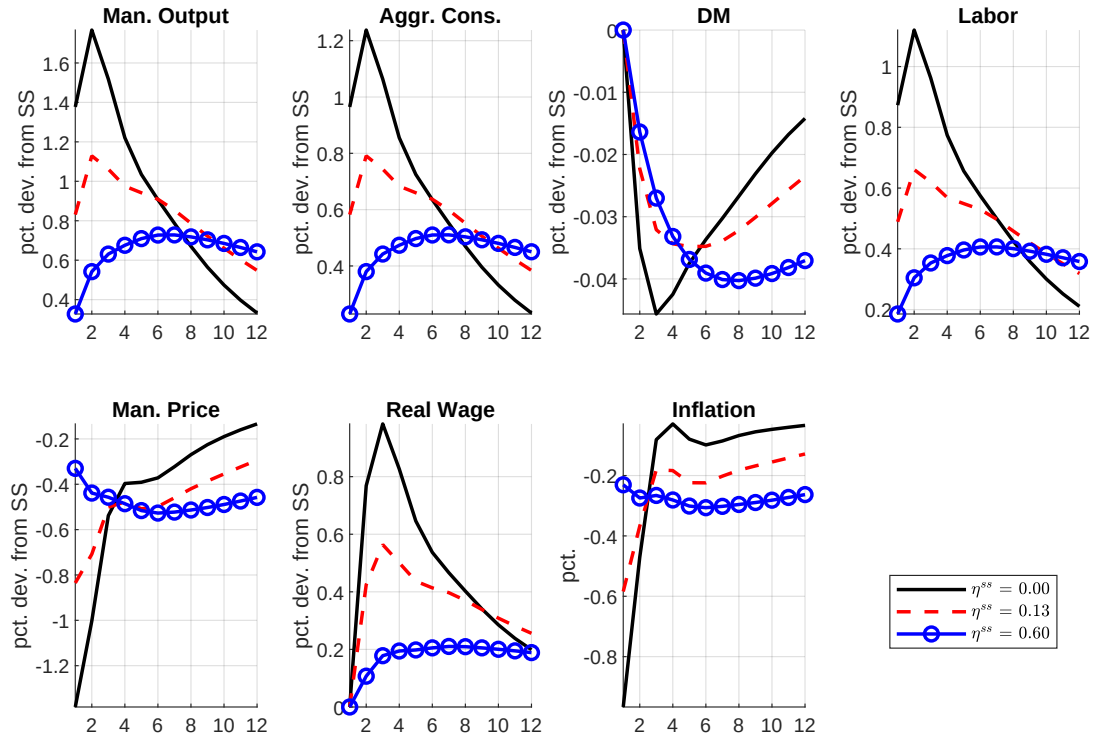
Note: This figure plots the decomposition of variations in output and the distribution margin attributable to all shocks within the estimated model.

Figure 4. Responses to a Money Supply Shock



Note: Responses to a 10% increase in the real money supply. The scale for all subfigures, except for inflation, is expressed as a percentage deviation from the steady state. We examine three scenarios of the steady-state distribution friction parameter $\eta^{ss} = 0, 0.13, 0.60$, corresponding to steady-state distribution margin values δ^{ss} of 0, 0.16, and 0.46, respectively. The horizontal axis measures the number of quarters following the shock.

Figure 5. Responses to an Interest Rate Shock under the Taylor Rule



Note: Responses to a 1% negative shock to the nominal interest rate. The scale for all subfigures, except for inflation, is expressed as a percentage deviation from the steady state. We investigate three scenarios of the steady-state distribution friction parameter $\eta^{ss} = 0, 0.13, 0.60$, corresponding to steady-state distribution margin values δ^{ss} of 0, 0.16, and 0.46, respectively. The horizontal axis denotes the number of quarters following the shock.

Table 1. Summary Statistics

Variable	Observations	Mean	Median	Std. Dev.	IQR
Distribution margin	101534	0.39%	-0.08%	10.71%	6.06%
GDP	91041	10.51%	10.34%	6.12%	7.44%
IP index	106331	0.00%	0.34%	2.06%	1.72%
CPI	106331	0.46%	0.44%	0.28%	0.35%
Construction cost	106331	0.51%	0.27%	1.25%	0.45%
Consumer confidence	98034	0.11%	0.31%	3.54%	4.56%
Exchange rate	106331	0.20%	0.08%	3.07%	3.49%
WTI oil price	106700	0.16%	0.25%	5.16%	5.69%
FTSE index	106700	0.24%	0.33%	3.30%	3.98%

Note: All variables are presented as growth rates and are observed at mixed frequencies. The distribution margin is at the state level and recorded weekly. GDP is measured at the state level and reported annually. Industrial production (IP) is at the national level and observed monthly. CPI is measured at the national level and published monthly. Construction costs are at the state level and are observed monthly. Consumer confidence is at the national level and observed monthly. Exchange rate data is at the national level and measured monthly. WTI crude oil spot price data is recorded weekly. The FTSE stock market index is at the national level and is observed weekly. These variables are utilized in the benchmark regressions. "Std. Dev." represents standard deviation, and "IQR" represents the inter-quartile range.

Table 2. Cyclicalities of the Distribution Margin: Baseline Regression Results

	Dependent Variable: Distribution Margin Growth						
GDP Growth	-0.017*** (0.004)	-0.021*** (0.004)	-0.022*** (0.004)	-0.023*** (0.004)	-0.023*** (0.004)	-0.023*** (0.004)	-0.023*** (0.004)
Distribution Margin Growth (Lag)		-0.242*** (0.023)	-0.242*** (0.023)	-0.242*** (0.023)	-0.237*** (0.024)	-0.237*** (0.024)	-0.237*** (0.024)
IP Growth			0.020 (0.027)	0.019 (0.027)	0.022 (0.030)	0.019 (0.033)	0.019 (0.033)
Overall Inflation			-0.623*** (0.105)	-0.637*** (0.105)	-0.671*** (0.113)	-0.638*** (0.110)	-0.638*** (0.108)
Construction Costs Growth				0.043** (0.021)	0.053** (0.023)	0.051** (0.022)	0.051** (0.022)
Consumer Confidence (Growth)					-0.010 (0.015)	-0.006 (0.012)	-0.005 (0.013)
FX Rate (Growth)						-0.036** (0.017)	-0.036** (0.017)
Oil Price Growth						0.038*** (0.010)	0.038*** (0.011)
FTSE Brazil Growth							-0.001 (0.014)
Constant	0.002 (0.003)	0.002 (0.003)	0.007* (0.003)	0.006* (0.003)	0.008** (0.004)	0.008** (0.004)	0.008** (0.004)
Week Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	53,914	53,604	53,604	53,604	50,776	50,776	50,776
Adj-R-Sqr	0.005	0.064	0.065	0.065	0.062	0.062	0.062

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the state level for all regressions. Fuel products, including gasoline, diesel, ethanol, and others, are pooled in the regressions. Regression results for individual fuel types align with the pooled case and can be provided upon request.

Table 3. Estimated Parameters

Parameters	Prior Mean	Post. Mean	90 pct HPD interval		Prior	Post. Std.
β	0.990	0.994	0.989	0.997	Beta	0.002
θ	5.000	5.039	4.642	5.381	Normal	0.208
φ	0.670	0.600	0.598	0.604	Normal	0.003
$\bar{\eta}$	0.130	0.039	0.004	0.091	Normal	0.029
χ	1.500	2.191	1.342	2.549	Normal	0.393
γ	0.700	0.886	0.809	0.947	Normal	0.044
ρ_Z^M	0.700	0.900	0.888	0.907	Normal	0.009
ρ_Z^S	0.700	0.899	0.897	0.900	Normal	0.001
ρ_m	0.700	0.709	0.700	0.717	Normal	0.006
Standard Deviation of Shocks						
Shocks	Prior Mean	Post. Mean	90 pct HPD interval		Prior	Post. Std.
ε_Z^M	0.010	0.049	0.044	0.055	Inv. Gamma	0.004
ε_Z^S	0.010	0.169	0.061	0.295	Inv. Gamma	0.019
ε_m	1.000	1.105	0.796	1.862	Inv. Gamma	0.284

Note: The model is estimated using three series: the growth rate of the manufacturing sector, real GDP growth rate, and the inflation rate. All data are sourced from the St. Louis Federal Reserve Economic Data (FRED) and have been seasonally adjusted. A 90% Highest Posterior Density (HPD) interval is the range of values within a posterior distribution that contains 90% of the total probability mass, such that every point inside the interval has a higher probability density than any point outside it. In other words, it means the interval is centered around the most probable values of the parameter.

Table 4. Difference in Gains from Reducing Distribution Friction across Two Models

Horizons (Quarters)	$C_{M,t}$	C_t	L_t	$P_{M,t}$	w_t	π_t
0	4.95	4.95	10.25	4.92	-62.73	4.93
2	4.50	4.49	11.41	5.47	3.62	5.92
4	4.92	4.92	12.60	8.31	4.06	17.60
6	4.64	4.63	12.14	5.08	3.92	5.24
8	4.65	4.65	11.90	6.08	3.69	6.94
10	4.80	4.80	12.34	5.91	4.06	6.59
12	4.64	4.63	12.03	5.48	3.82	5.86

Note: This table presents the difference in gains from responses to a 10% increase in the real money supply, when transitioning from $\eta^{ss} = 0.13$ to $\eta^{ss} = 0$, between the model featuring state-dependent distribution friction and one with time-invariant friction. Essentially, it shows the difference in gains between the two models, expressed as $g_h^{DID}(x) = g_h^{SD}(x) - g_h^{TI}(x)$ as discussed in the main text. All values are expressed as percentage gains relative to the baseline scenario with $\eta^{ss} = 0.13$.

Appendix Accompanying “State-dependent Distribution Friction and the Transmission of Monetary Policy”

Dong Cheng and Nam T. Vu

A. Data

A.1. Brazilian Fuel Price Data

The fuel price data are from the *National Agency of Petroleum, Natural Gas and Biofuels* (ANP in Portuguese). Specifically, the data are from the *Fuel Price Survey*, a weekly price survey of stations across states from 2004 to 2019. The data contain weekly information on average station prices and distribution costs for each state, along with the standard deviation of prices across stations within each state. To get a sense of the coverage of the dataset by state, Table A.1 presents the distribution margins by state for all products, along with the descriptive statistics when averaging across all states (i.e., Total). Similarly, Fig. A.1 provides a heat map of the constructed distribution margins by states when pooling the whole sample (first column), for the beginning of the sample in 2004 (second column), and for the last year of the sample in 2019 (third column).

A.1.1. Sensitivity Analysis: Brazilian State-level Composition

The state-level compositional concern, however, is conceptually distinct from a trend-cycle concern: HP filtering rebalances the time-series component but does not absorb slow-moving, state-specific dynamics such as distributor entry into structurally growing states. To address this directly, we augment Eqn. (22) with state-specific linear and quadratic time trends; results are reported in Appendix Table A.11. The GDP-growth coefficient remains negative across all specifications, but it attenuates and is no longer statistically distinguishable from zero in the most controlled specification once the lagged distribution margin is included with the full vector of macro controls. We interpret this transparently: a non-trivial part of the unconditional counter-

cyclical correlation in our panel reflects state-specific trends consistent with a compositional story, although its sign remains negative in every specification.

A.1.2. Robustness Check: Using Binary CODACE Recession Indicator

As an additional robustness check, we replace state-level GDP growth with a binary CODACE recession indicator constructed from the official monthly chronology of the Brazilian economic cycle dating committee at FGV/IBRE; results are in Appendix Table A.12. The recession dummy enters with a positive sign in every specification and is statistically significant at the 1% level in the most-controlled specification without state trends, indicating that the distribution margin grows faster, on average, in CODACE-dated recessions than in expansions and corroborating the model's prediction of a counter-cyclical distribution margin.

A.1.3. Robustness Check: Subsample Analysis

We further exploit the Brazilian fuel data to address the concern that the cyclical correlation between distribution margins and GDP could partly reflect counter-cyclical Petrobras pricing rather than the structural friction emphasized by the model. Specifically, we drop sub-periods of well-documented Petrobras intervention, namely the Rousseff-era subsidy regime (2011:M1–2014:M12) and the post-trucker-strike federal diesel subsidy (2018:M5–2018:M6), and we separately restrict the sample to years of relatively stable international oil prices, defined as years with within-year coefficient of variation of weekly WTI below the cross-year median. Appendix Tables A.13 and A.14 show that the GDP-growth coefficient remains negative, statistically significant at the 1% level, and quantitatively very close to the benchmark estimate in both robustness samples.

A.2. Validating the Micro Evidence for Counter-cyclical Distribution Margin

As discussed earlier, the empirical insights from Brazilian microdata on fuel prices should have significant implications for the broader economy. In this subsection, we further extend our main empirical finding on the countercyclicality of the distribution margin to other economies and

sectors. These validation exercises help reinforce the robustness and general applicability of the countercyclicality result.

A.3. Alternative Evidence from U.S. Farm Food Products

We first collect data on distribution margins from U.S. historical archives, which provide an extensive time series for the farm food sector. Previous studies, such as [Burstein et al. \(2003\)](#), have used input-output tables to compute distribution margins, but these typically provide only snapshots for a single year or a few years. In contrast, we have directly recorded distribution margins from a reliable historical archive spanning nearly 40 years.

Specifically, we gather data from the table *Series, T 352-355, Farm-to-Retail Price Spreads of Farm Food Products: 1913-1957*, compiled in the *Historical Statistics of the United States, Colonial Times to 1957*. This table presents the share of farm value in the retail cost of the market basket of farm food products, defined as the ratio of the producer price to the retail price. Thus, we can calculate the distribution margin as one minus this share. Additionally, we obtain real GDP per capita data from the 2020 Maddison Project Database. The U.S. historical distribution margin data span from 1913 to 1952, with growth rates available from 1914 to 1952.

Fig. [A.3](#) illustrates the correlation between the annual growth rates of the distribution margin and real GDP per capita. A strong negative association between the two growth rates indicates significant countercyclicality of the distribution margin. Specifically, the correlation coefficient between the growth rates of the distribution margin and real GDP per capita is -0.66, statistically significant at the 1% level. This finding confirms the countercyclicality observed in high-frequency Brazilian microdata on fuel prices. Moreover, cross-validation suggests that this countercyclicality extends across sectors, economies, and periods. Cross-validation of countercyclical distribution margins between Brazilian microdata and U.S. historical data confirms the benchmark model's prediction in **Proposition 1**, highlighting its representativeness and pervasiveness across the data.

[Figure [A.3](#) here]

It is worth noting that cross-validation using historical U.S. farm product data particularly

helps alleviate concerns that the distributional margin of fuel, previously constrained to lower bounds in studies such as [Crucini and Landry \(2019\)](#), could bias estimates of countercyclicality obtained from Brazilian fuel price data, rendering them unrepresentative. The average distribution margin from the U.S. historical data is 56%, which is at the upper bounds in [Crucini and Landry \(2019\)](#) and is of similar magnitude to those found in [Burstein et al. \(2003\)](#).

A.4. Alternative Evidence from Indian Manufacturing

We complement our empirical findings on the counter-cyclical distribution margin in the Brazilian fuel sector using further evidence from India. To do so, we leverage India's Annual Survey of Industries (ASI) from the Ministry of Statistics and Programme Implementation (MoSPI). The dataset contains microdata at the firm and product levels, including distribution costs, sales, and taxes, among others, for the period 1998 to 2023.

Recall that, following [Peter and Ruane \(2022\)](#), we construct a distribution margin for firm i that sells a set of products J_i as follows

$$DM_{it} = \frac{1}{J_i} \sum_{j=1}^{J_i} \frac{\text{Distribution Cost}_{ijt}}{\text{Sale}_{ijt} - \text{Sale Tax and Excise Duty}_{ijt}} \quad (\text{A.1})$$

where $\text{Distribution Cost}_{ijt}$ denotes annual distribution cost by firm i for product j , Sale_{ijt} is the gross sale value for product j by firm i , and $\text{Sale Tax and Excise Duty}_{ijt}$ includes all sale tax and excise duty incurred by firm i for product j . Given the heterogeneity in the products each firm has, we compute the average distribution margin as in Eqn. (A.1). Data on key Indian macroeconomic aggregates (GDP growth rates, inflation, the exchange rate, and oil prices) are sourced from the Federal Reserve Bank of St. Louis's FRED.

Upon constructing the firm-level distribution margin measure, we consider the following regression

$$g_{i,t}^{DM} = \beta_0 + \beta_1 g_t^{GDP} + \theta \mathbb{X}_t + \Gamma_1 \text{State}_s + \Gamma_2 \text{Firm}_i + v_{i,t}, \quad (\text{A.2})$$

where $g_{i,t}^{DM}$ is growth rate of the distribution margin defined according to Eqn. (A.1), g_t^{GDP} is

the growth rate of Indian GDP, $\mathbb{X}_t$ denotes a set of other macroeconomic aggregates (the growth rates of inflation, exchange rate, and oil prices), State_s and Firm_i denotes state and firm fixed effects, respectively.

Table A.5 presents the estimates from Eqn. (A.2), showing that distribution margin growth is counter-cyclical. This result dovetails with our empirical findings on the countercyclicality of distribution margins in the Brazilian data. A substantial proportion of firms in the ASI dataset have distribution margin growth rates for only one year, making lagged distribution margin growth infeasible for these firms. We therefore exclude it in the benchmark specifications in Table A.5. Nevertheless, when the lagged term is added in the final column of Table A.5, the distribution margin remains counter-cyclical.

[Table A.5 here]

It is worth emphasizing that cross-validating our findings with large-scale Indian manufacturing survey data significantly mitigates concerns about relying solely on the Brazilian fuel sector to assess the cyclical behavior of the distribution margin. Brazil is both a major commodity producer and a country in which fuel prices are heavily influenced by the state-run enterprise Petrobras, raising questions about the representativeness of fuel-sector dynamics. Moreover, fuel and farm-food markets constitute relatively small shares of overall economic activity. In contrast, manufacturing is one of the largest sectors in India, making it a highly reasonable setting for evaluating whether the countercyclical distribution margin is sensitive to sector choice.

However, we also note that Indian manufacturing data are available only at an annual frequency, which limits their ability to capture high-frequency cyclical variation. For this reason, we retain the Brazilian data as the benchmark and use the Indian evidence as indispensable complementary cross-validation. In addition, in the Indian case, we have more limited access to high-quality macroeconomic controls, such as state-level GDP, construction cost indices, and consumer confidence measures.

ASI Data: Sensivity Analysis Because the ASI is an unbalanced panel, the unconditional

counter-cyclicalities could partly reflect compositional shifts (for example, higher-margin marginal firms exiting in downturns mechanically raising the average margin). We therefore re-estimate Eqn. (A.2) on two balanced sub-panels of firms continuously present in every year of a fixed window: 2014-2023 (30,802 firms) and 2017-2023 (35,171 firms). Results are reported in Appendix Tables A.15 and A.16. In the more recent 2017-2023 balanced sub-panel, the GDP-growth coefficient is -0.030 and statistically significant at the 1% level in the most-controlled specification, somewhat stronger than the unbalanced benchmark of -0.011 . In the longer 2014-2023 window, the coefficient on GDP growth is essentially zero, although the sign on the lagged distribution margin and the standard cyclical signs on exchange rate and oil prices are preserved. We read these results as confirming that the Indian countercyclicalities are robust in the recent balanced sub-panel, while also acknowledging that part of the long-window unconditional pattern is associated with firm entry and exit.

B. The Augmented Model: Real-term Equilibrium Conditions

For ease of computation, we normalize all variables in the system by the price index in the previous period, P_{t-1} . The equilibrium conditions solve for the real-term series of

$$\left\{ \begin{array}{l} C_t, w_t, L_t, i_t, l_t, m_t, \mu_t, \pi_t^p, p_{M,t}, p_{M,t}^c, p_{S,t}, \\ C_{M,t}, C_{S,t}, \Pi_t, t_t, b_t, \delta_t, a_{1t}, a_{2t}, mc_t, U'(C_t) \end{array} \right\}. \text{ The full set of equilibrium equations is as follows:}$$

The pricing block

$$p_{S,t} = \left(\frac{\theta}{\theta - 1} \right) \left(\frac{w_t}{Z_{S,t}} \right) \quad (\text{A.3})$$

$$p_{M,t}^c = \frac{\theta}{\theta - 1} \frac{a_{1t}}{a_{2t}} \quad (\text{A.4})$$

$$p_{M,t}^{1-\theta} = \varphi \left(p_{M,t}^c \right)^{1-\theta} + (1 - \varphi) p_{M,t-1}^{1-\theta} \frac{1}{(\pi_{t-1}^p)^{1-\theta}} \quad (\text{A.5})$$

$$a_{1t} = U'(C_t) m c_t^R p_{M,t}^\theta C_{M,t} + (\beta \varphi) E_t a_{1t+1} (\pi_t^p)^{\theta+1} \quad (\text{A.6})$$

$$a_{2t} = U'(C_t) p_{M,t}^{\theta-1} C_{M,t} + (\beta \varphi) E_t a_{2t+1} (\pi_t^p)^{\theta-1} \quad (\text{A.7})$$

$$\pi_t^p = p_{M,t}^\gamma p_{S,t}^{1-\gamma} \quad (\text{A.8})$$

$$m c_t^R = \left(1 + \frac{\eta_t}{\theta - 1} \frac{Z_{M,t}}{Z_{S,t}} \right) \frac{w_t}{Z_{M,t}} + \eta_t \frac{w_t}{Z_{S,t}} \quad (\text{A.9})$$

Here we are defining two auxiliary equations

$$\begin{aligned} a_{1t} &= \frac{A_{1t}}{p_{t-1}^{\theta+1}} \\ a_{2t} &= \frac{A_{2t}}{p_{t-1}^{\theta-1}}, \end{aligned}$$

and hence a straightforward derivation yields

$$\begin{aligned} a_{1t} &= U'(C_t) m c_t^R p_{M,t}^\theta C_{M,t} + (\beta \varphi) E_t a_{1t+1} (\pi_t^p)^{\theta+1} \\ a_{2t} &= U'(C_t) p_{M,t}^{\theta-1} C_{M,t} + (\beta \varphi) E_t a_{2t+1} (\pi_t^p)^{\theta-1} \end{aligned}$$

The distribution friction function

$$\eta_t = \bar{\eta} \times \left(1 + \xi \frac{p_{S,t} C_{S,t}}{C_{t-1}} \right)^\iota \quad (\text{A.10})$$

The rest of the equilibrium conditions

$$U'(C_t) = \frac{1}{C_t} \quad (\text{A.11})$$

$$w_t = \frac{\phi}{\phi - 1} \frac{E_{t-1} \kappa L_t}{E_{t-1} L_t / \mu_t} \quad (\text{A.12})$$

$$w_t l_t = \left(\frac{\theta - 1}{\theta} \right) p_{S,t} (C_{S,t} + \eta_t C_{M,t}) + \frac{m c_t^R}{p_{M,t}} (p_{M,t} - \eta_t p_{S,t}) C_{M,t} \quad (\text{A.13})$$

$$\mu_t = \pi_t^p C_t \quad (\text{A.14})$$

$$\pi_t^p C_t = \frac{p_{M,t} C_{M,t}}{\gamma} \quad (\text{A.15})$$

$$\pi_t^p C_t = \frac{p_{S,t} C_{S,t}}{1 - \gamma} \quad (\text{A.16})$$

$$\frac{1}{1 + i_{t+1}} = E_t \left\{ \beta \frac{\mu_t}{\mu_{t+1}} \frac{1}{\pi_t^p} \right\} \quad (\text{A.17})$$

$$\frac{m_t}{\pi_t^R} = \chi \frac{1 + i_{t+1}}{i_{t+1}} C_t \quad (\text{A.18})$$

$$\delta_t = \frac{\eta_t \frac{Z_{M,t}}{Z_{S,t}}}{1 + \eta_t \frac{\theta}{\theta - 1} \frac{Z_{M,t}}{Z_{S,t}}} \quad (\text{A.19})$$

$$p_{M,t} C_{M,t} + p_{S,t} C_{S,t} + m_t + b_t = \frac{m_{t-1}}{\pi_{t-1}^p} + (1 + i_t) \frac{b_{t-1}}{\pi_{t-1}^p} + w_t L_t + \Pi_t - t_t \quad (\text{A.20})$$

$$\Pi_t = p_{S,t} C_{S,t} + p_{M,t} C_{M,t} - w_t l_t \quad (\text{A.21})$$

$$L_t = l_t \quad (\text{A.22})$$

$$m_t - \frac{m_{t-1}}{\pi_{t-1}^p} = t_t \quad (\text{A.23})$$

The exogenous shock processes

$$\log(Z_{M,t}) = \rho_Z^M \log(Z_{M,t-1}) + \varepsilon_{M,t}, \quad (\text{A.24})$$

$$\log(Z_{S,t}) = \rho_Z^S \log(Z_{S,t-1}) + \varepsilon_{S,t}, \quad (\text{A.25})$$

$$\log(m_t) = (1 - \rho_m) \log(\bar{m}) + \rho_m \log(m_{t-1}) + \varepsilon_{m,t}, \quad (\text{A.26})$$

C. Estimating the Quantitative Model

Apart from ϕ , κ , ξ , and ι , we estimate the rest of the parameters using Bayesian methods à la [Smets and Wouters \(2007\)](#) to data on the manufacturing sector's growth rate, real GDP growth, and the inflation rate. All data are seasonally adjusted and aligned with their model counterparts, which yields a total of 96 quarterly observations.

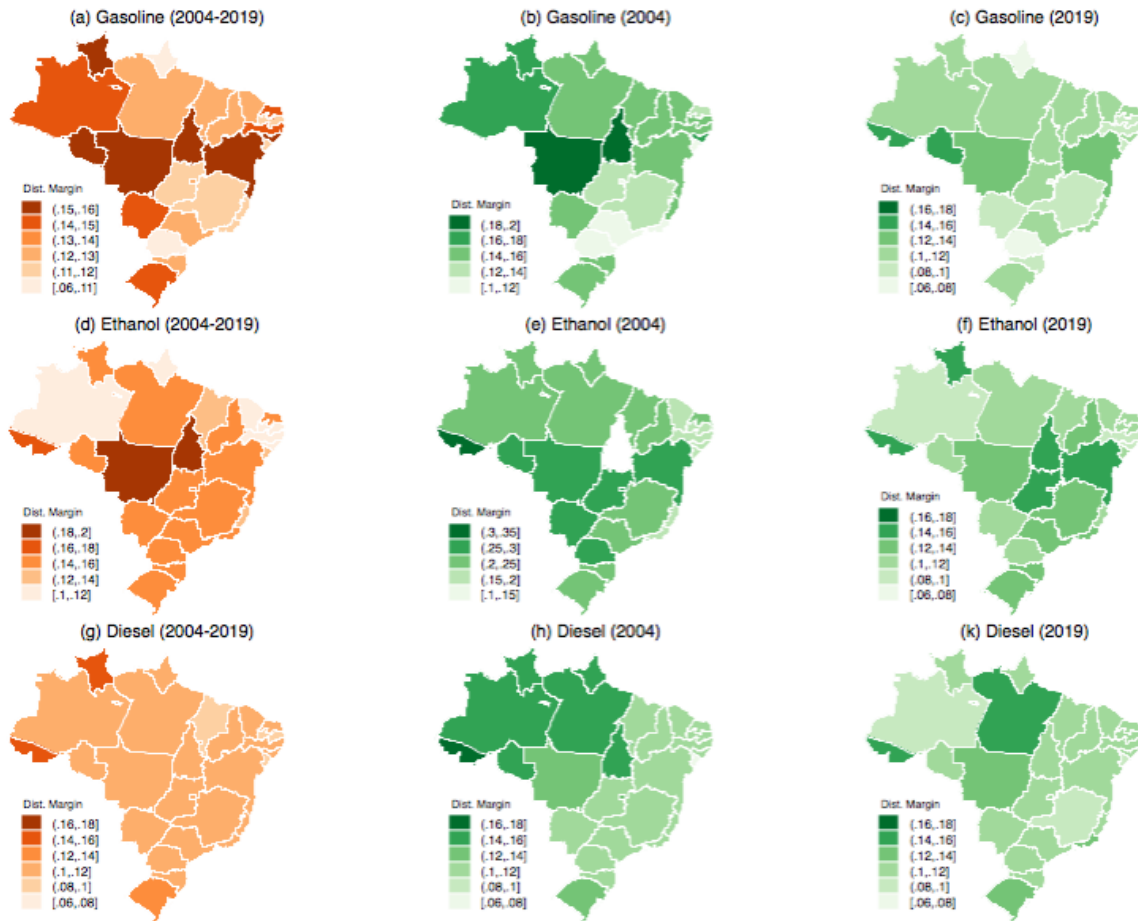
Specifically, we follow [Smets and Wouters \(2007\)](#) and adopt a prior value of 0.99 for the subjective discount factor β . Turning to the Calvo parameter φ , we set the prior mean at 0.67, implying an average price duration of roughly three quarters. Such a calibration choice follows from and aligns with the extensive evidence on micro-price adjustments (see, e.g., [Bils and Klenow, 2004](#)). Next, we assign a prior mean of 5 to the elasticity of substitution θ to be consistent with estimates from [Hsieh and Klenow \(2009\)](#). For the distribution friction size parameter $\bar{\eta}$, we choose a prior mean of 0.13, which delivers a steady state distribution margin δ^{SS} in line with our Brazilian microdata estimate of 16 percent. For the weight on real balance χ and the manufacturing expenditure share γ , we adopt priors with means of 1.5 and 0.7, respectively, following [Corsetti and Dedola \(2005\)](#). We also assign relatively uninformative priors to the persistence parameters governing the productivity and money supply processes, centered at 0.7. Such a choice allows the data to determine the degree of persistence while remaining agnostic *ex ante* about whether the shocks are highly persistent or relatively transitory.

Turning to the numerical implementation, we use the standard Kalman filter algorithm to evaluate the likelihood. In particular, we employ the Random Walk Metropolis-Hastings (RWMH) algorithm to characterize the posterior distribution of the parameters. To ensure a global search for the posterior mode, we use a Monte Carlo-based optimization routine. We discipline our estimation by generating two blocks of 10,000 replications each. In other words, we draw 20,000 simulations from the posterior, using a jumping scale of 0.8 to ensure the chains achieve an appropriate acceptance rate.

C.1. Sensitivity Analysis for ξ

We conduct further sensitivity analysis with respect to the state-dependence parameter ξ . First, when we re-estimate the augmented model with ξ added to the list of estimated parameters under a diffuse Normal prior $N(1, 0.5)$, the marginal posterior is centered close to one but is moderately wide, consistent with the limited information that the three aggregate observable series carry about the extent of state-dependency in the distribution friction function (Appendix Table A.17). Second, we discipline ξ externally by matching the model-implied OLS slope of distribution-margin growth on output growth, simulated under alternative values of ξ , to the Brazilian empirical slope of approximately -0.023 from Table 2. The model-implied slope is monotonically decreasing in ξ and exceeds the empirical target in absolute value at every grid point considered, with the closest match at the lower bound of the grid (Appendix Table A.18 and Appendix Figure A.8). Thus, our calibration of $\xi = 1$ is conservative in the sense that it overstates the model-implied slope relative to the data; reducing ξ toward zero brings the model closer to the empirical target, and the qualitative results of the paper are preserved throughout.

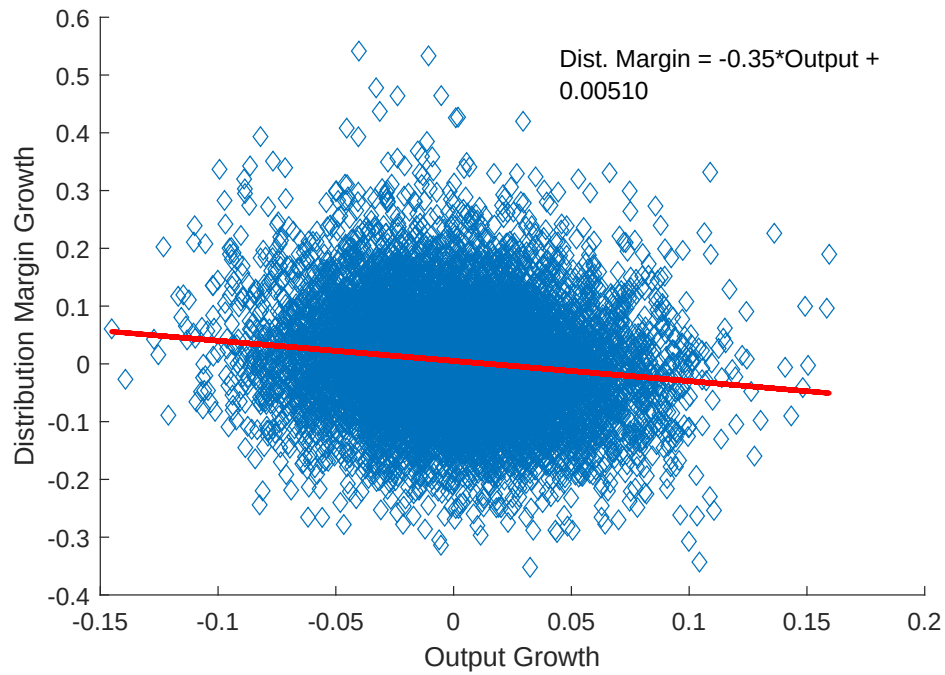
Figure A.1. Spatial Distribution of the Distribution Margin



Note: The distribution margin is the ratio of distribution cost to retail price. The three subgraphs in the left column depict the average distribution margin from 2004 to 2019, while the middle and right columns display the distribution margins for 2004 and 2019, respectively.

Figure A.2. Cyclicity of the Distribution Margin: Augmented Models

Panel A: With State-dependent Distribution Friction

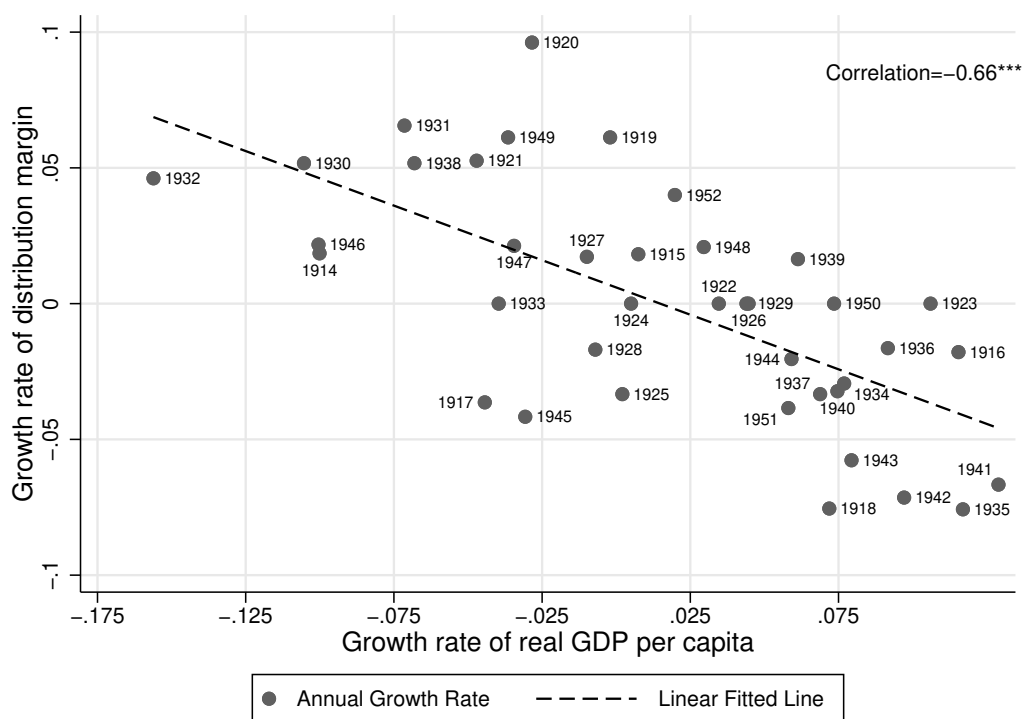


Panel B: Without State-dependent Distribution Friction



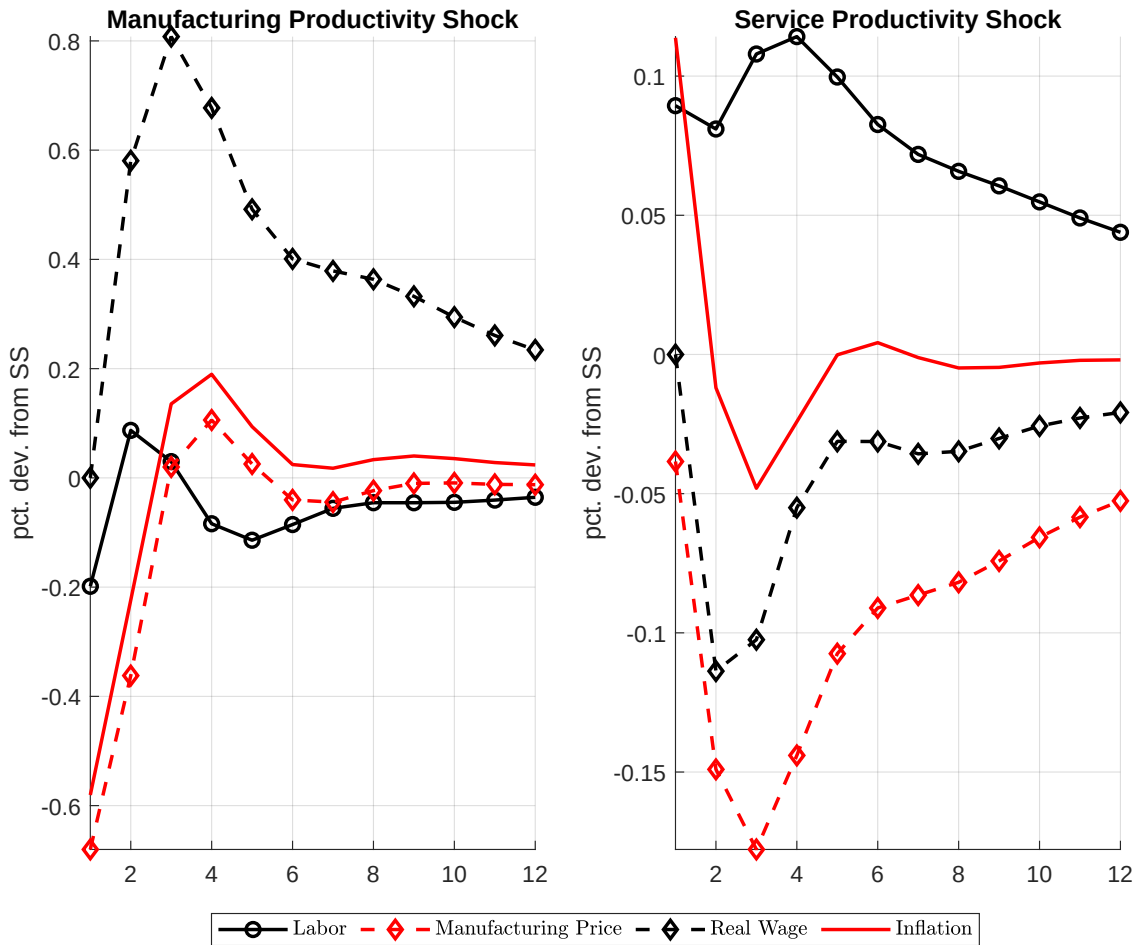
Note: Each scatter plot depicts the growth rates of simulated output and the distribution margin over 10,000 iterations, employing parameter values corresponding to the alternative model specifications. All shocks are included. The solid line represents the Ordinary Least Squares (OLS) linear fit, with all coefficients statistically significant at the 5% level. Panel A presents our benchmark quantitative model, whereas Panel B presents an alternative version of the model without state dependence in the distribution friction.

Figure A.3. Cyclicalty of the Distribution Margin: Evidence for Cross-validation



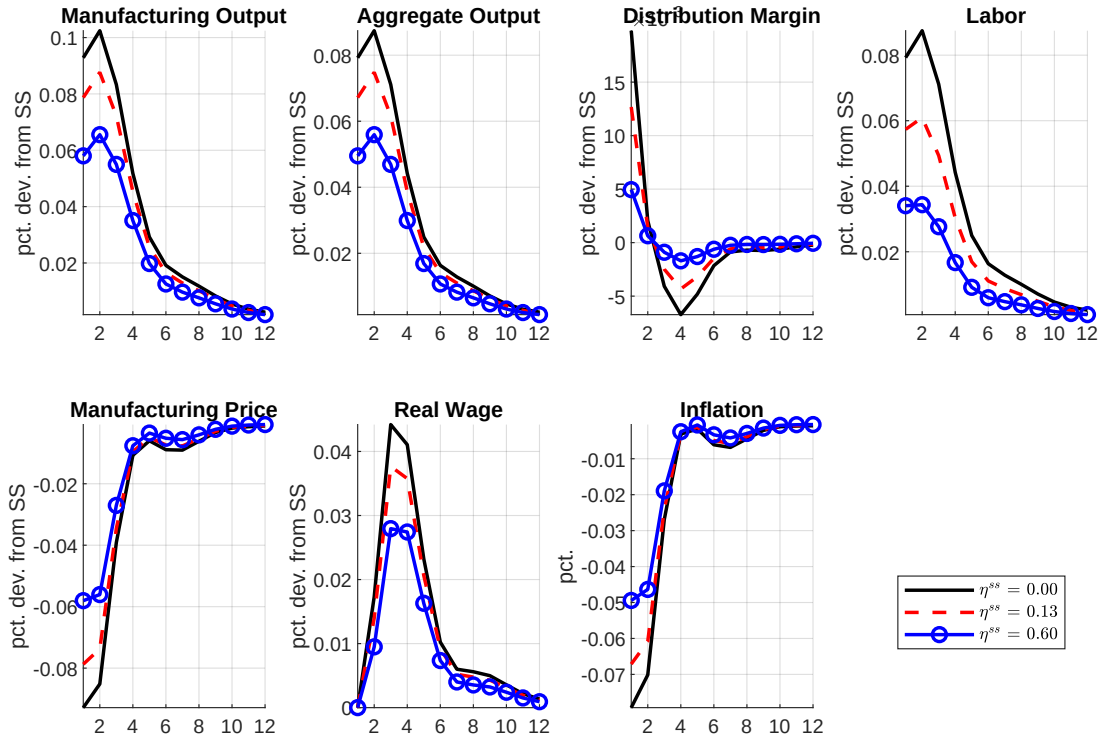
Note: A scatter plot displays the annual growth rates of the distribution margin and real GDP per capita, utilizing U.S. historical data spanning from 1914 to 1952. The dataset is compiled from the *Historical Statistics of the United States, Colonial Times to 1957* for distribution margin data and the Maddison Project Database 2020 for real GDP per capita data.

Figure A.4. Responses to Sector-specific Productivity Shocks



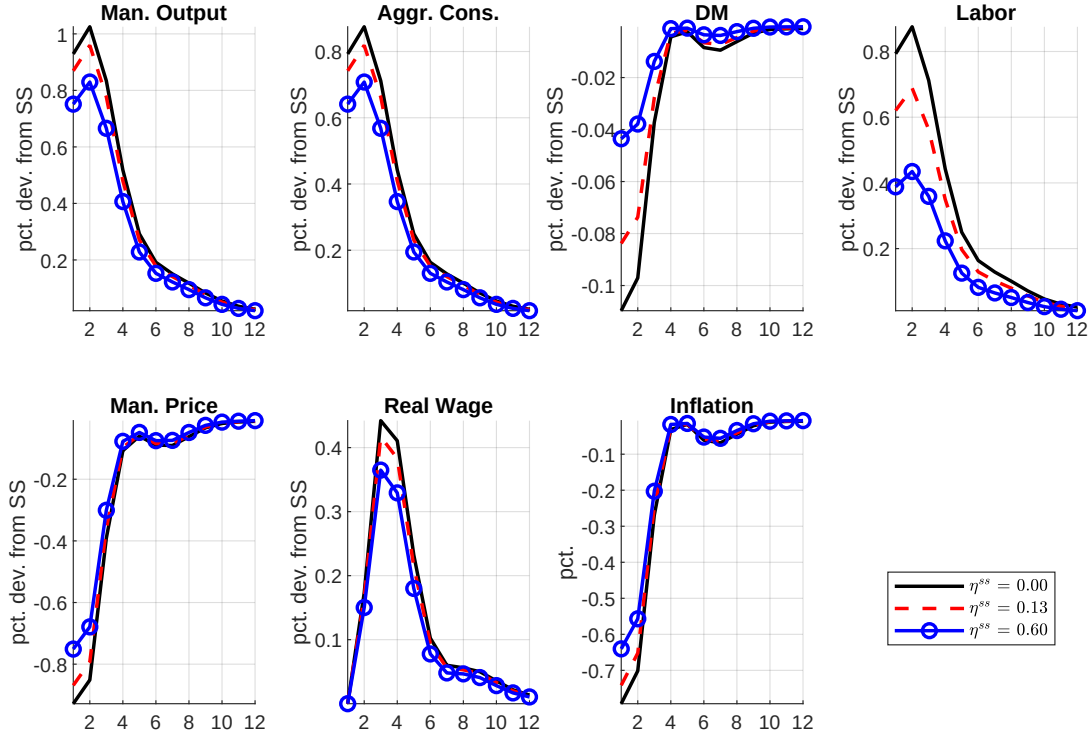
Note: Responses to a 1% increase in manufacturing productivity and a 1% decrease in service productivity. The scale for all subfigures represents the percentage deviation from the steady state. The horizontal axis measures the number of quarters following the shock.

Figure A.5. Responses to a Money Supply Shock: Alternative Frictional Form of η_t



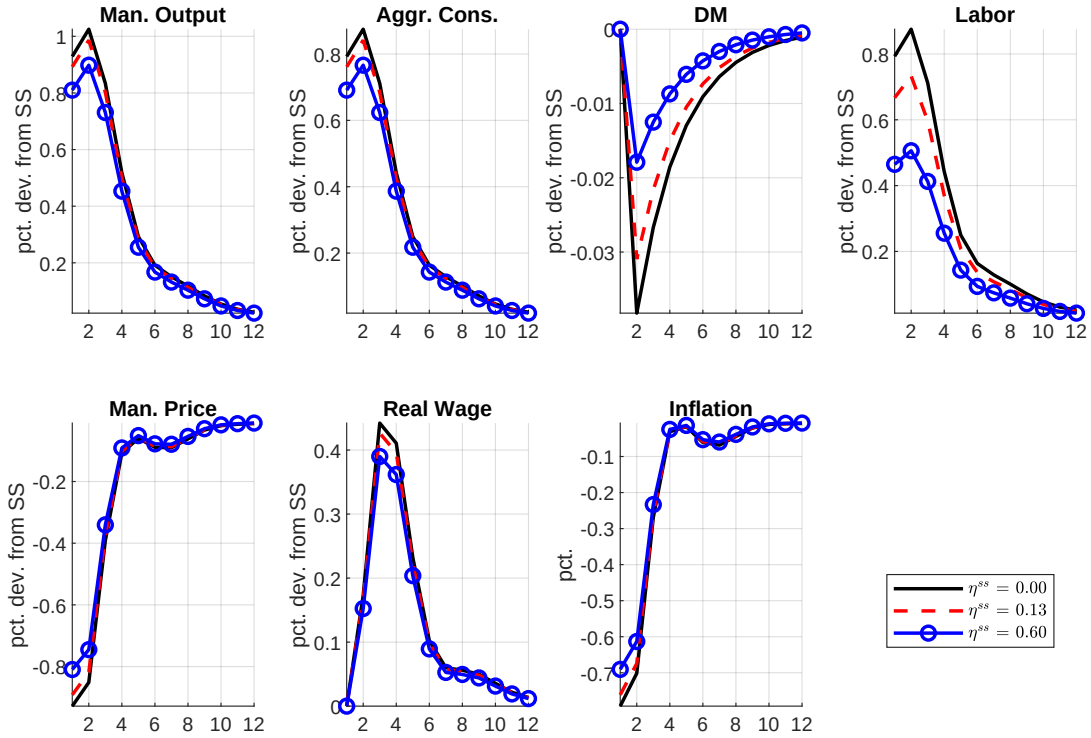
Note: Responses to a 10% increase in the real money supply. The scale for all subfigures reflects the percentage deviation from the steady state, except for inflation. We investigate three scenarios of steady-state distribution friction: $\eta^{ss} = 0, 0.13, 0.60$, corresponding to steady-state distribution margins δ^{ss} of 0, 0.16, and 0.46, respectively. The horizontal axis represents the number of quarters following the shock. The alternative functional form of the distribution friction is given by $\eta_t = \bar{\eta} \times \left(1 + \xi \frac{C_{S,t}}{C_{t-1}}\right)^t$.

Figure A.6. Responses to a Money Supply Shock: Current Expenditure Specification



Note: Responses to a 10% increase in the real money supply. The scale for all subfigures, except for inflation, is expressed as a percentage deviation from the steady state. We examine three scenarios of the steady-state distribution friction parameter $\eta^{ss} = 0, 0.13, 0.60$, corresponding to steady-state distribution margin values δ^{ss} of 0, 0.16, and 0.46, respectively. The horizontal axis measures the number of quarters following the shock. To allow the distribution friction to depend on current expenditures solely, we re-specify the friction function as $\eta_t = \bar{\eta} \times \left(1 + \xi \frac{P_{S,t} C_{S,t}}{P_t C_t}\right)^l$.

Figure A.7. Responses to a Money Supply Shock: Specification with Service TFP



Note: Responses to a 10% increase in the real money supply. The scale for all subfigures, except for inflation, is expressed as a percentage deviation from the steady state. We examine three scenarios of the steady-state distribution friction parameter $\eta^{ss} = 0, 0.13, 0.60$, corresponding to steady-state distribution margin values δ^{ss} of 0, 0.16, and 0.46, respectively. The horizontal axis measures the number of quarters following the shock. To allow the distribution friction to depend on service productivity, we re-specify the friction function as $\eta_t = \bar{\eta} \times \left(1 + \xi \frac{1}{Z_{S,t}} \frac{P_{S,t} C_{S,t}}{P_{t-1} C_{t-1}}\right)^t$.

Table A.1. Distribution Margins by States

	Mean	SD	Min	Max	Obs
Acre	0.16	0.03	-0.00	0.44	2348
Alagoas	0.13	0.03	0.02	0.23	2353
Amapa	0.11	0.03	0.01	0.32	2028
Amazonas	0.12	0.03	0.04	0.35	2355
Bahia	0.14	0.04	0.07	0.34	2355
Ceara	0.12	0.02	0.07	0.24	2355
Distrito Federal	0.13	0.04	0.04	0.36	2328
Espirito Santo	0.12	0.02	0.03	0.26	2352
Goiias	0.13	0.04	0.07	0.37	2352
Maranhao	0.12	0.04	0.03	0.41	2352
Mato Grosso	0.15	0.04	0.09	0.33	2355
Mato Grosso do Sul	0.13	0.04	0.06	0.35	2355
Minas Gerais	0.12	0.03	0.07	0.32	2355
Para	0.13	0.02	0.06	0.31	2352
Paraiba	0.11	0.03	0.05	0.29	2354
Parana	0.12	0.03	0.07	0.30	2355
Pernambuco	0.12	0.03	0.04	0.24	2353
Piaui	0.12	0.03	0.02	0.31	2346
Rio Grande do Norte	0.13	0.03	0.06	0.29	2347
Rio Grande do Sul	0.14	0.02	0.07	0.29	2354
Rio de Janeiro	0.13	0.02	0.08	0.28	2355
Rondonia	0.14	0.03	0.01	0.35	2335
Roraima	0.15	0.03	0.04	0.31	2337
Santa Catarina	0.13	0.03	0.05	0.34	2352
Sao Paulo	0.14	0.03	0.08	0.28	2355
Sergipe	0.11	0.03	0.04	0.25	2355
Tocantins	0.16	0.06	0.05	0.46	2352
Total	0.13	0.04	-0.00	0.46	63145

Note: This table presents key descriptive statistics for distribution margins for each state across all fuel products in the benchmark Brazilian fuel price sample. Total is a simple average for each statistic across all states.

Table A.2. Cyclicalities of the Distribution Margin: Regression using HP-filtered Data

	Dependent Variable: Distribution Margin Growth (HP-filtered)							
GDP Growth (HP-filtered)	-0.048** (0.016)	-0.020** (0.009)	-0.023** (0.009)	-0.024*** (0.008)	-0.026*** (0.008)	-0.025*** (0.008)	-0.025*** (0.008)	-0.030*** (0.008)
Distribution Margin Growth (Lag, HP-filtered)		0.649*** (0.033)	0.647*** (0.033)	0.648*** (0.034)	0.647*** (0.034)	0.647*** (0.034)	0.649*** (0.034)	0.646*** (0.034)
Industrial Production (HP-filtered)			0.021*** (0.007)	0.030** (0.013)	0.033** (0.013)	0.035** (0.013)	0.027* (0.013)	0.016 (0.014)
Overall Inflation (HP-filtered)				0.025 (0.039)	0.016 (0.040)	0.011 (0.039)	0.011 (0.039)	-0.037 (0.044)
Construction Costs (HP-filtered)					0.033*** (0.009)	0.034*** (0.009)	0.036*** (0.009)	0.026*** (0.009)
Consumer Confidence (HP-filtered)						-0.002 (0.003)	-0.006* (0.003)	-0.017*** (0.004)
Oil Price (HP-filtered)							0.004*** (0.001)	0.002* (0.001)
FX Rate (HP-filtered)								0.014*** (0.004)
Constant	0.000** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000** (0.000)
Product FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,446	6,338	6,338	6,338	6,338	6,338	6,338	6,338
Adj-R-Sqr	0.002	0.420	0.421	0.421	0.422	0.422	0.423	0.424

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the state level for all regressions. Fuel products, including gasoline, diesel, ethanol, and others, are pooled in the regressions. Regression results for individual fuel types align with the pooled case and can be provided upon request. All variables are at the monthly frequency and HP-filtered using their natural logs.

Table A.3. Calibrated Parameters

Parameters	Value	Notes	Sources
ϕ	3	Elasticity of substitution across different types of labor	Behar (2025) ; Havranek et al. (2024)
κ	1	Disutility associated with labor	Kaplan and Schulhofer-Wohl (2018)
ι	0.5	Concavity of the distribution friction function	Jermann and Quadrini (2012)
ξ	1	Degree of state dependence of the distribution friction	Author's choice

Table A.4. Cyclicalities of the Distribution Margin: Manufacturing versus Services

	Dependent Variable: Distribution Margin Growth						
Manufacturing Growth	-0.001 (0.012)	0.000 (0.017)	-0.002 (0.016)	-0.003 (0.016)	-0.003 (0.017)	-0.003 (0.017)	-0.004 (0.017)
Service Growth	-0.065*** (0.020)	-0.084*** (0.027)	-0.097*** (0.027)	-0.098*** (0.027)	-0.099*** (0.027)	-0.097*** (0.027)	-0.097*** (0.027)
GDP Growth	-0.001 (0.007)	0.000 (0.008)	0.003 (0.008)	0.003 (0.008)	0.004 (0.009)	0.003 (0.009)	0.003 (0.009)
Distribution Margins (Lag)		-0.242*** (0.023)	-0.242*** (0.023)	-0.242*** (0.023)	-0.237*** (0.024)	-0.237*** (0.024)	-0.237*** (0.024)
IP Growth			0.027 (0.026)	0.026 (0.026)	0.029 (0.029)	0.026 (0.032)	0.026 (0.033)
Overall Inflation			-0.803*** (0.113)	-0.821*** (0.113)	-0.861*** (0.122)	-0.828*** (0.119)	-0.830*** (0.117)
Construction Costs Growth				0.049** (0.020)	0.058** (0.022)	0.056** (0.022)	0.056** (0.022)
Consumer Confidence (Growth)					-0.010 (0.015)	-0.007 (0.013)	-0.007 (0.013)
FX Rate (Growth)						-0.031* (0.017)	-0.031* (0.016)
Oil Price Growth						0.038*** (0.010)	0.038*** (0.011)
FTSE Brazil Growth							-0.002 (0.014)
Constant	0.007* (0.004)	0.009* (0.004)	0.015*** (0.004)	0.015*** (0.004)	0.017*** (0.005)	0.017*** (0.005)	0.017*** (0.005)
Week Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	53,914	53,604	53,604	53,604	50,776	50,776	50,776
Adj-R-Sqr	0.005	0.065	0.065	0.065	0.063	0.063	0.063

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the state level for all regressions. Fuel products, including gasoline, diesel, ethanol, and others, are pooled in the regressions. Regression results for individual fuel types are consistent with those from the pooled case and can be provided upon request.

Table A.5. Cyclicalities of the Distribution Margin: Indian Manufacturing Data

	Dependent Variable: Distribution Margin Growth				
	(1)	(2)	(3)	(4)	(5)
GDP Growth	-0.0149*** (0.00406)	-0.0149*** (0.00406)	-0.0123*** (0.00414)	-0.0151*** (0.00432)	-0.0105** (0.00531)
Inflation		-0.0143** (0.00625)	-0.0206*** (0.00670)	-0.0191*** (0.00674)	-0.0149 (0.0108)
FX Growth			0.00929*** (0.00339)	0.0116*** (0.00352)	0.00714* (0.00411)
Oil Price Growth				0.00114** (0.000519)	0.00142** (0.000637)
Distribution Margin Growth (Lag)					-0.490*** (0.0291)
Constant	0.00661*** (0.000232)	0.00745*** (0.000437)	0.00746*** (0.000437)	0.00739*** (0.000439)	0.00664*** (0.000653)
Firm FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Observations	385,199	385,199	385,199	385,199	207,555
Adj- R^2	0.143	0.143	0.143	0.143	0.350

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Alternative price data from India to re-affirm the countercyclicality of the distribution margin. Robust standard errors are in parentheses. Details of the variable and sample construct are provided in the accompanying Appendix.

Table A.6. Estimated Model Comparison

Model Variants/Data	Data	Model: Preferred	Model: Variant I	Model: Variant II
$\text{Var}(Y_t) \times 10^{-4}$	1.47	4.29	4.21	3.80
$\text{Var}(C_t) \times 10^{-4}$	5.54	5.47	5.62	5.27
$\text{Var}(\pi_t) \times 10^{-4}$	8.47	4.69	4.63	4.22
$\text{Corr}(Y_t, Y_{t-1})$	0.23	0.32	0.32	0.31
$\text{Corr}(C_t, C_{t-1})$	0.32	0.32	0.34	0.33
$\text{Corr}(\pi_t, \pi_{t-1})$	0.51	0.39	0.39	0.37
$\text{Corr}(\pi_t, Y_t)$	-0.09	-0.95	-0.96	-0.96
$\text{Corr}(\pi_t, C_t)$	-0.01	-0.93	-0.93	-0.92
$\text{Corr}(Y_t, C_t)$	0.78	0.98	0.97	0.94
Posterior Model Probability	-	0.56	0.29	0.15

Note: This table shows selected statistics for the preferred estimated model against two model variants. Under all models, we calibrate the curvature parameter $\iota = 0.5$ following [Jermann and Quadrini \(2012\)](#), the elasticity of substitution parameter $\phi = 3$ following [Behar \(2025\)](#), and the labor disutility parameter $\kappa = 1$ following [Kaplan and Schulhofer-Wohl \(2018\)](#). The preferred model incorporates state-dependency in the distribution friction ($\xi = 1$) and estimates the parameter $\bar{\eta}$ that governs the size of the state-dependent distribution friction, along with other model parameters. Variant I is estimated without state-dependency (i.e., $\xi = 0$) in the distribution friction and with the exogenous steady-state distribution friction ($\bar{\eta}$) estimated along with other parameters in the model. Variant II is estimated without state-dependency in the distribution friction and with the steady-state distribution friction ($\bar{\eta}$) calibrated. Posterior model probability represents the likelihood that each model is the true data-generating process, given the observed data and the prior probabilities assigned to the models. Hence, a higher posterior model probability means that the model is more supported by the data.

Discussion: In Table A.6, we present evidence that our preferred quantitative model better aligns with the data than alternative models, as it exhibits the highest posterior model probability. Additionally, it consistently shows smaller deviations in the second-order moments of consumption and inflation variances. While the preferred model does not consistently outperform the alternatives in matching autocorrelations, it excels at capturing correlations among key macroeconomic variables.

Table A.7. Estimated Parameters Robustness ($\iota = 0.4$)

Parameters	Prior Mean	Post. Mean	90 pct HPD interval		Prior	Post. Std.
β	0.990	0.995	0.989	0.997	Beta	0.003
θ	5.000	4.939	4.600	5.024	Normal	0.146
φ	0.670	0.602	0.599	0.609	Normal	0.005
η	0.180	0.035	0.017	0.090	Normal	0.029
χ	1.500	2.185	1.783	2.637	Normal	0.235
γ	0.700	0.862	0.795	0.886	Normal	0.033
ρ_Z^M	0.700	0.900	0.883	0.910	Normal	0.012
ρ_Z^S	0.700	0.898	0.896	0.900	Normal	0.002
ρ_m	0.700	0.710	0.700	0.718	Normal	0.007
Standard Deviation of Shocks						
Shocks	Prior Mean	Post. Mean	90 pct HPD interval		Prior	Post. Std.
ε_Z^M	0.010	0.050	0.044	0.058	Inv. Gamma	0.005
ε_Z^S	0.010	0.123	0.071	0.151	Inv. Gamma	0.023
ε_m	1.000	1.053	0.876	1.384	Inv. Gamma	0.186

Note: This is to check the robustness of estimated parameters in our preferred model when we change the calibration value of the curvature parameter ι from 0.5 to 0.4. The model is estimated using three series: the growth rate of the manufacturing sector, the real GDP growth rate, and the inflation rate. All data are sourced from the St. Louis Federal Reserve Economic Data (FRED) and have been seasonally adjusted. A 90% Highest Posterior Density (HPD) interval is the range of values within a posterior distribution that contains 90% of the total probability mass, such that every point inside the interval has a higher probability density than any point outside it. In other words, it means the interval is centered on the parameter's most probable value.

Table A.8. Estimated Parameters Robustness ($\iota = 0.6$)

Parameters	Prior Mean	Post. Mean	90 pct HPD interval		Prior	Post. Std.
β	0.990	0.989	0.988	0.989	Beta	<0.001
θ	5.000	5.051	5.036	5.064	Normal	0.008
φ	0.670	0.602	0.600	0.605	Normal	0.002
η	0.130	0.046	0.021	0.087	Normal	0.020
χ	1.500	1.533	1.427	1.633	Normal	0.062
γ	0.700	0.891	0.876	0.907	Normal	0.009
ρ_Z^M	0.700	0.890	0.878	0.900	Normal	0.008
ρ_Z^S	0.700	0.897	0.892	0.900	Normal	0.003
ρ_m	0.700	0.871	0.867	0.875	Normal	0.003
Standard Deviation of Shocks						
Shocks	Prior Mean	Post. Mean	90 pct HPD interval		Prior	Post. Std.
ε_Z^M	0.010	0.053	0.045	0.061	Inv. Gamma	0.005
ε_Z^S	0.010	0.173	0.147	0.196	Inv. Gamma	0.016
ε_m	1.000	1.637	1.518	1.758	Inv. Gamma	0.072

Note: This is to check the robustness of estimated parameters in our preferred model when we change the calibration value of the curvature parameter ι from 0.5 to 0.6. The model is estimated using three series: the growth rate of the manufacturing sector, the real GDP growth rate, and the inflation rate. All data are sourced from the St. Louis Federal Reserve Economic Data (FRED) and have been seasonally adjusted. A 90% Highest Posterior Density (HPD) interval is the range of values within a posterior distribution that contains 90% of the total probability mass, such that every point inside the interval has a higher probability density than any point outside it. In other words, it means the interval is centered on the parameter's most probable value.

Table A.9. Estimated Parameters Robustness ($\xi = 0.5$)

Parameters	Prior Mean	Post. Mean	90 pct HPD interval		Prior	Post. Std.
β	0.990	0.994	0.988	0.997	Beta	0.004
θ	5.000	5.053	4.779	5.213	Normal	0.181
φ	0.670	0.600	0.598	0.604	Normal	0.003
η	0.130	0.035	0.007	0.091	Normal	0.036
χ	1.500	2.437	1.914	2.705	Normal	0.243
γ	0.700	0.872	0.811	0.898	Normal	0.036
ρ_Z^M	0.700	0.902	0.886	0.912	Normal	0.012
ρ_Z^S	0.700	0.898	0.895	0.900	Normal	0.002
ρ_m	0.700	0.709	0.700	0.716	Normal	0.007
Standard Deviation of Shocks						
Shocks	Prior Mean	Post. Mean	90 pct HPD interval		Prior	Post. Std.
ε_Z^M	0.010	0.049	0.042	0.054	Inv. Gamma	0.003
ε_Z^S	0.010	0.128	0.081	0.161	Inv. Gamma	0.023
ε_m	1.000	0.771	0.446	1.240	Inv. Gamma	0.351

Note: This is to check the robustness of estimated parameters in our preferred model when we change the calibration value of the state-dependence degree parameter ξ from 1 to 0.5. The model is estimated using three series: the growth rate of the manufacturing sector, the real GDP growth rate, and the inflation rate. All data are sourced from the St. Louis Federal Reserve Economic Data (FRED) and have been seasonally adjusted. A 90% Highest Posterior Density (HPD) interval is the range of values within a posterior distribution that contains 90% of the total probability mass, such that every point inside the interval has a higher probability density than any point outside it. In other words, it means the interval is centered on the parameter's most probable value.

Table A.10. Estimated Parameters Robustness ($\xi = 1.5$)

Parameters	Prior Mean	Post. Mean	90 pct HPD interval		Prior	Post. Std.
β	0.990	0.994	0.991	0.997	Beta	0.001
θ	5.000	5.412	5.174	5.504	Normal	0.111
φ	0.670	0.601	0.599	0.604	Normal	0.002
η	0.130	0.034	0.003	0.086	Normal	0.038
χ	1.500	1.972	1.850	2.231	Normal	0.111
γ	0.700	0.832	0.745	0.884	Normal	0.058
ρ_Z^M	0.700	0.899	0.888	0.906	Normal	0.009
ρ_Z^S	0.700	0.899	0.897	0.901	Normal	0.002
ρ_m	0.700	0.713	0.701	0.718	Normal	0.008
Standard Deviation of Shocks						
Shocks	Prior Mean	Post. Mean	90 pct HPD interval		Prior	Post. Std.
ε_Z^M	0.010	0.049	0.045	0.058	Inv. Gamma	0.004
ε_Z^S	0.010	0.108	0.064	0.136	Inv. Gamma	0.031
ε_m	1.000	1.126	0.953	1.446	Inv. Gamma	0.218

Note: This is to check the robustness of estimated parameters in our preferred model when we change the calibration value of the state-dependence degree parameter ξ from 1 to 1.5. The model is estimated using three series: the growth rate of the manufacturing sector, the real GDP growth rate, and the inflation rate. All data are sourced from the St. Louis Federal Reserve Economic Data (FRED) and have been seasonally adjusted. A 90% Highest Posterior Density (HPD) interval is the range of values within a posterior distribution that contains 90% of the total probability mass, such that every point inside the interval has a higher probability density than any point outside it. In other words, it means the interval is centered on the parameter's most probable value.

Table A.11. Cyclicalty of the Distribution Margin: State-specific Linear and Quadratic Trends

	Dependent Variable: Distribution Margin (Growth)					
GDP Growth	-0.009** (0.004)	-0.008** (0.003)	-0.007* (0.004)	-0.004 (0.005)	-0.002 (0.005)	-0.004 (0.004)
Distribution Margins (Lag)		-0.242*** (0.023)	-0.238*** (0.024)		-0.243*** (0.023)	-0.238*** (0.024)
IP Growth			0.026 (0.033)			0.023 (0.034)
Overall Inflation			-0.828*** (0.119)			-0.741*** (0.131)
Construction Costs Growth			0.060** (0.023)			0.057** (0.023)
Consumer Confidence (Growth)			-0.005 (0.013)			-0.007 (0.013)
FX Rate (Growth)			-0.029 (0.017)			-0.030* (0.017)
Oil Price Growth			0.038*** (0.011)			0.038*** (0.011)
FTSE Brazil Growth			-0.000 (0.015)			-0.001 (0.015)
Constant	-0.002 (0.003)	-0.002 (0.003)	0.003 (0.004)	-0.005 (0.003)	-0.008** (0.003)	-0.004 (0.004)
Week Dummies	Yes	Yes	Yes	Yes	Yes	Yes
State Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Product Dummies	Yes	Yes	Yes	Yes	Yes	Yes
State Linear Trends	Yes	Yes	Yes	Yes	Yes	Yes
State Quadratic Trends	No	No	No	Yes	Yes	Yes
Observations	53,914	53,604	50,776	53,914	53,604	50,776
Adj-R-Sqr	0.005	0.065	0.063	0.005	0.065	0.063

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Robustness for the benchmark Brazilian regression in Eqn. (22) augmenting the specification with state-specific linear and state-specific quadratic time trends (in columns 4-6). Standard errors are clustered at the state level. Fuel products (gasoline, diesel, ethanol, and others) are pooled. Trends are constructed as years from the start of the sample (2004).

Table A.12. Cyclicity of the Distribution Margin: Binary CODACE Recession Indicator

	Dependent Variable: Distribution Margin (Growth)			
CODACE Recession	0.001 (0.001)	0.001* (0.001)	0.002** (0.001)	0.000 (0.001)
Distribution Margins (Lag)		-0.228** (0.021)	-0.225** (0.023)	-0.227** (0.023)
IP Growth			-0.075* (0.042)	-0.077* (0.042)
Overall Inflation			-1.116** (0.169)	-0.962** (0.198)
Construction Costs Growth			0.033 (0.020)	0.052** (0.021)
Consumer Confidence (Growth)			0.021 (0.013)	0.018 (0.013)
FX Rate (Growth)			-0.046** (0.017)	-0.038** (0.017)
Oil Price Growth			0.033** (0.011)	0.033** (0.011)
FTSE Brazil Growth			-0.016 (0.015)	-0.019 (0.015)
Constant	0.001 (0.004)	0.001 (0.004)	0.010** (0.004)	-0.003 (0.004)
Week Dummies	Yes	Yes	Yes	Yes
State Dummies	Yes	Yes	Yes	Yes
Product Dummies	Yes	Yes	Yes	Yes
State Linear Trends	No	No	No	Yes
State Quadratic Trends	No	No	No	Yes
Observations	62,697	62,275	56,855	56,855
Adj-R-Sqr	0.005	0.058	0.057	0.058

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Robustness for the benchmark Brazilian regression replacing the state-level GDP growth rate with the binary CODACE monthly recession indicator from the official chronology of the Brazilian economic cycle dating committee at FGV/IBRE. Within our 2004:M5-2019:M6 sample, the indicator equals one during 2008:M7-2009:M1 and 2014:M3-2016:M12. A positive coefficient on the recession dummy is consistent with a counter-cyclical distribution margin. Column 4 also adds state-specific linear and quadratic time trends. Standard errors are clustered at the state level.

Table A.13. Cyclicity of the Distribution Margin: Dropping Petrobras Intervention Sub-periods

	Dependent Variable: Distribution Margin (Growth)		
GDP Growth	-0.020*** (0.005)	-0.023*** (0.005)	-0.025*** (0.005)
Distribution Margins (Lag)		-0.250*** (0.030)	-0.245*** (0.032)
IP Growth			0.027 (0.036)
Overall Inflation			-0.584*** (0.130)
Construction Costs Growth			-0.002 (0.069)
Consumer Confidence (Growth)			0.016 (0.014)
FX Rate (Growth)			-0.065*** (0.022)
Oil Price Growth			0.047*** (0.011)
FTSE Brazil Growth			-0.001 (0.018)
Constant	0.000 (0.004)	0.001 (0.004)	0.007 (0.004)
Week Dummies	Yes	Yes	Yes
State Dummies	Yes	Yes	Yes
Product Dummies	Yes	Yes	Yes
Observations	37,057	36,758	33,930
Adj-R-Sqr	0.007	0.070	0.067

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Robustness for the benchmark Brazilian regression dropping sub-periods of well-documented Petrobras intervention: the Rousseff-era subsidy regime (2011:M1-2014:M12), during which Petrobras held domestic gasoline and diesel prices below international parity, and the post-trucker-strike federal diesel subsidy (2018:M5-2018:M6). Standard errors are clustered at the state level.

Table A.14. Cyclicalities of the Distribution Margin: Years of Stable International Oil Prices

	Dependent Variable: Distribution Margin (Growth)		
GDP Growth	-0.022*** (0.005)	-0.029*** (0.007)	-0.024*** (0.007)
Distribution Margins (Lag)		-0.280*** (0.017)	-0.276*** (0.019)
IP Growth			0.105 (0.062)
Overall Inflation			-1.131*** (0.231)
Construction Costs Growth			0.077** (0.034)
Consumer Confidence (Growth)			-0.009 (0.019)
FX Rate (Growth)			0.001 (0.022)
Oil Price Growth			-0.024 (0.021)
FTSE Brazil Growth			-0.008 (0.030)
Constant	0.007* (0.004)	0.007* (0.004)	0.018*** (0.005)
Week Dummies	Yes	Yes	Yes
State Dummies	Yes	Yes	Yes
Product Dummies	Yes	Yes	Yes
Observations	29,243	29,121	26,293
Adj-R-Sqr	0.010	0.089	0.086

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Robustness for the benchmark Brazilian regression restricted to years of relatively stable international oil prices, defined as years with within-year coefficient of variation of weekly WTI crude oil prices below the cross-year median. Standard errors are clustered at the state level.

Table A.15. Cyclicalities of the Indian Distribution Margin: Balanced Sub-panel 2014-2023

	Dependent Variable: Distribution Margin Growth (Balanced 2014-2023)				
	(1)	(2)	(3)	(4)	(5)
GDP Growth	0.000232 (0.00603)	-0.0000119 (0.00610)	-0.000514 (0.00618)	-0.000490 (0.00661)	0.00431 (0.00759)
Inflation		-0.00599 (0.0228)	0.00217 (0.0284)	0.00214 (0.0285)	0.0376 (0.0320)
FX Growth			-0.00555 (0.0119)	-0.00558 (0.0126)	-0.0124 (0.0136)
Oil Price Growth				-0.0000820 (0.000912)	0.000689 (0.00104)
Distribution Margin Growth (lag)					-0.551*** (0.0154)
Constant	0.00663*** (0.000371)	0.00696*** (0.00128)	0.00674*** (0.00135)	0.00675*** (0.00135)	0.00493*** (0.00149)
Observations	84,511	84,511	84,511	84,511	56,538
Adj-R-Sqr	-0.182	-0.182	-0.182	-0.182	0.110

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Robustness for the Indian ASI regression in Eqn. (A.2) on a balanced sub-panel of firms continuously present in every year from 2014 to 2023 (30,802 firms). Robust standard errors are reported in parentheses. The reported adjusted R^2 corresponds to the within-firm-state-FE specification estimated with `reghdfe`, in which the absorbed firm and state fixed effects soak up most of the variation in distribution margin growth; the resulting adjusted R^2 on the residualized sample can take small or negative values without affecting the interpretation of the slope coefficients.

Table A.16. Cyclicalities of the Indian Distribution Margin: Balanced Sub-panel 2017-2023

	Dependent Variable: Distribution Margin Growth (Balanced 2017-2023)				
	(1)	(2)	(3)	(4)	(5)
GDP Growth	-0.00680 (0.00633)	-0.00800 (0.00642)	-0.0163** (0.00691)	-0.0313** (0.00898)	-0.0301** (0.00993)
Inflation		-0.0439 (0.0378)	-0.0431 (0.0378)	-0.00781 (0.0407)	0.0384 (0.0417)
FX Growth			-0.0489** (0.0167)	-0.0375** (0.0176)	-0.0473** (0.0185)
Oil Price Growth				0.00378** (0.00156)	0.00494** (0.00160)
Distribution Margin Growth (lag)					-0.569** (0.0184)
Constant	0.00637** (0.000383)	0.00887** (0.00218)	0.0111** (0.00233)	0.00899** (0.00251)	0.00682** (0.00255)
Observations	64,332	64,332	64,332	64,332	43,726
Adj-R-Sqr	-0.255	-0.255	-0.255	-0.255	0.080

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Robustness for the Indian ASI regression on a balanced sub-panel of firms continuously present in every year from 2017 to 2023 (35,171 firms). Robust standard errors are reported in parentheses. As in Table A.15, the reported adjusted R^2 is computed on the residualized sample after absorbing firm and state fixed effects with `reghdfe`, and can take small or negative values without affecting the interpretation of the slope coefficients.

Table A.17. Posterior of ξ under a Diffuse Prior $N(1, 0.5)$

Parameters	Prior Mean	Post. Mean	90 pct HPD interval		Prior	Post. Std.
β	0.990	0.993	0.988	0.997	Beta	0.003
θ	5.000	5.082	4.937	5.257	Normal	0.100
φ	0.670	0.602	0.600	0.605	Normal	0.002
η	0.180	0.101	0.060	0.143	Normal	0.026
ξ	1.000	0.997	0.575	1.359	Normal	0.258
χ	1.500	2.011	1.433	2.602	Normal	0.376
γ	0.700	0.824	0.774	0.886	Normal	0.034
ρ_H	0.700	0.894	0.884	0.900	Normal	0.006
ρ_N	0.700	0.898	0.896	0.900	Normal	0.002
ρ_m	0.700	0.709	0.700	0.722	Normal	0.009
Standard Deviation of Shocks						
Shocks	Prior Mean	Post. Mean	90 pct HPD interval		Prior	Post. Std.
ε_H^z	0.010	0.051	0.045	0.058	Inv. Gamma	0.004
ε_N^z	0.010	0.101	0.065	0.147	Inv. Gamma	0.025
ε_m	1.000	1.315	0.951	1.705	Inv. Gamma	0.245

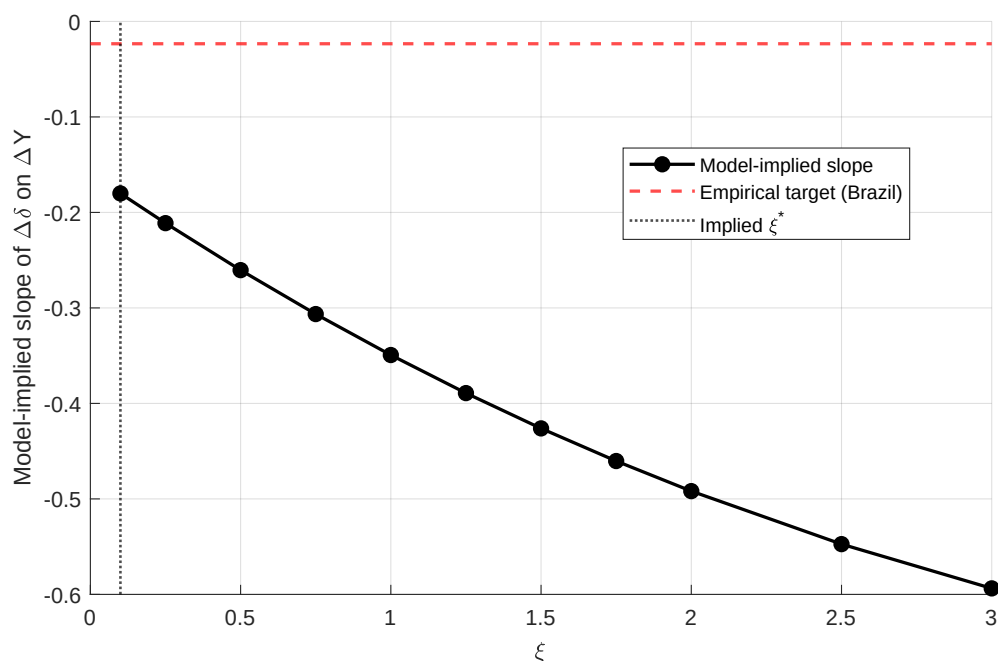
Note: Bayesian estimation of the augmented model treating the state-dependence parameter ξ as an additional estimated parameter under a diffuse Normal prior $N(1, 0.5)$. The remaining priors and the three observable series (manufacturing growth, real GDP growth, inflation) are identical to the benchmark specification. A 90% Highest Posterior Density (HPD) interval is the range of values that contains 90% of the posterior mass and is centered on the parameter's most probable value. The marginal posterior of ξ is centered close to one but moderately wide, consistent with limited identification of ξ from aggregate data.

Table A.18. Disciplining ξ from the Brazilian Empirical Slope

ξ	0.10	0.25	0.50	0.75	1.00	1.25	1.50	1.75	2.00	2.50	3.00
Model-implied slope	-0.180	-0.211	-0.260	-0.306	-0.349	-0.389	-0.426	-0.460	-0.492	-0.547	-0.594
<i>Empirical Brazilian slope (Table 2): -0.023</i>											
<i>Implied ξ^* matching the empirical slope: 0.10</i>											

Note: For each value of ξ on the grid, the augmented model is simulated for 10,000 quarters (after a 1,000-quarter burn-in) at the estimated benchmark parameters and the OLS slope of the growth rate of δ_t on the growth rate of aggregate consumption is reported. The empirical target slope of -0.023 is read from the most-controlled benchmark Brazilian regression in Table 2. The implied ξ^* is the grid value that minimizes the absolute distance between the model-implied and empirical slopes.

Figure A.8. Disciplining ξ from the Brazilian Empirical Slope



Note: Mapping from ξ to the model-implied OLS slope of distribution-margin growth on output growth, simulated under the augmented model at the benchmark estimated parameters. The dashed horizontal line is the Brazilian empirical target of approximately -0.023 from Table 2. The dotted vertical line marks the implied ξ^* .